Q-CRAFT Explorer Companion Guide

A course in climate-aware fiscal projection

Teal Insights & NatureFinance

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Preface

What Q-CRAFT is

Q-CRAFT is the IMF Fiscal Affairs Department's tool for projecting public finances under climate warming. The full name is the Quantitative Climate Risk Assessment Fiscal Tool. It takes a country's macroeconomic and fiscal data, adds assumptions about demography, productivity, prices and fiscal policy, and projects the debt-to-GDP ratio to 2099 under a baseline and six warming scenarios.

The Fiscal Affairs Department built it for its own climate technical assistance, and that is where it is used. The September 2023 mission to Uganda is one example: a five-day workshop with ministry staff, written up as *Uganda: PFM Climate Assessment* (IMF, 2024). Ministries then carry the results into their own documents. Uganda's Fiscal Risk Statement for FY 2024/25 has a chapter on climate fiscal risks whose figures are sourced to "QCRAFT (2023)".

This course teaches the tool through **Q-CRAFT Explorer**, an open-source Python reimplementation of the IMF's Excel workbook. Teal Insights and NatureFinance develop it and the code is MIT licensed. The economics is the IMF's. The web interface, the automatic data loading and the record of what you assumed are ours.

The questions it answers

Q-CRAFT answers a narrow set of questions, and it answers each one by comparing two runs of the same model. Four of them, in the register they usually arrive in:

- If warming follows the high scenario, where is our debt ratio in 2050? And in 2099?
- How much of that path is climate damage, as against where the debt ratio was heading anyway?
- Does climate damage on its own take us through the ceiling in our fiscal rule, and in which year?
- Which assumption moves the answer most: the demographic outlook, the interest rate, or how far spending can adjust when growth disappoints?

None of those is a forecast. Each one is a difference between two projections built on assumptions you chose, which is why the assumptions are worth arguing about and worth writing down.

Who this is for

This course is for anyone who has to run, interact with, or otherwise understand this class of fiscal projection tool. Ministry of finance economists come first, because they run the tool and then defend the numbers to the officials who sign them off. IFI and technical assistance staff come second, because they build capacity around tools like this one and a teaching sequence is easier to hand over than a workshop. Researchers and analysts come third, because they read the output and have to decide what it licenses them to say.

Where this course stands

This guide teaches what the tool does, why it exists, and what it leaves out. The third part belongs with the first two.

Practitioners disagree about Q-CRAFT, and some judge its damage estimates conservative. The tool's own documentation gives them their grounds. The scenarios model the slow effect of temperature on productivity, and they exclude natural disasters, sea-level rise, tipping points and the public spending that adaptation takes (User Guide, pp. 5-6). Under those channels the projected fiscal impact reads as a lower bound rather than as a central estimate of total impact. Chapter 6 sets this out with the sources.

Knowing a model's limits is part of knowing the model. This is the first in a series of guides to the models behind sovereign climate-fiscal analysis, and each one takes the same three questions: what the model is for, what it computes, and where it stops.

What you will be able to do

Three objectives, weighted equally:

1. **Run the tool.** Select a country, set the parameters, read the charts, export the results.
2. **Understand what you are doing when you run it.** One equation, three inputs, and a clear view of which of your assumptions is moving the answer.
3. **Interpret the output appropriately.** Say what a result supports and what it does not, with its strengths and its limits attached.

The first objective takes an afternoon. The second and third are the ones that hold up in a meeting.

How the course is organised

Seven modules, each built around something you can do at the end of it.

Module	What it gets you
0. How to use this course	The capstone stated up front, a self-assessment, and the path that fits your background
1. What Q-CRAFT does and why it exists	A full projection in ten minutes, and the map: three numbers, one equation
2. The debt equation	The debt identity rebuilt from words. Skippable if you use it already.

Module	What it gets you
3. Choosing the parameters	Every parameter set and defended, with the rationale written down
4. A worked example, end to end	One country from assumptions to a fiscal risk paragraph, then two more with less help
5. What the tool can and cannot tell you	The strengths, the documented exclusions, and which questions belong to a different tool
6. The capstone	The capstone brief and rubric, and a look back at where you started

The tool covers most of the world, so the examples move country by country and the mechanism picks the country. Demography reads clearest where the workforce is already shrinking, set against one where it is still growing fast. Productivity convergence reads clearest in an economy catching up to the frontier. The climate channel reads clearest where exposure is high. Module 4 runs one country the whole way through, and that country is Uganda, because it is the tool's golden-master verification country and every figure in the walkthrough can be checked against published documents. The exercises that follow it run on other countries, and the capstone runs on yours.

Start at Module 0. It takes about ten minutes and it will tell you which of the later modules you can skip.

The Glossary defines key terms and the References section gives annotated pointers to the primary sources underlying the methodology.

There is also an appendix on co-design and the SovTech vision. It sits outside the learning path on purpose: it is about the project behind the tool rather than about using it, and nothing in the capstone depends on it. Read it if you want to shape what V2 does.

What this course defers to

This is an educational companion to the IMF's User Guide (Tim and Rahman, 2024), which remains the authoritative methodology reference. Where this guide explains a concept, it cites the relevant User Guide section so you can go deeper.

> This is an initial version
>
> Q-CRAFT Explorer is a starting point rather than a finished product. We want to co-design the next version with you. See the co-design appendix for how to get involved.

> Try the App
>
> **Open Q-CRAFT Explorer** to follow along as you read. The source code is on GitHub.

Colophon

This course is open source under the MIT license, and it builds completely from what is in its repository. It sets in Inter, IBM Plex Serif and IBM Plex Mono, three faces under the SIL Open

Font License, bundled with the book rather than fetched from a font service. Clone the repository, run `quarto render docs/companion-guide`, and you get this book in the same three faces on a machine with no network access. Opening a page fetches no font from anyone's server, which matters in a ministry network that blocks outside hosts.

Teal Insights also publishes a house edition of the same content, set in licensed Klim Type Foundry faces where the license permits. Those font files are not in the repository, because the license that covers them covers tealinsights.com. Where they are absent, every font stack falls through to the open faces and the book still sets. No word, number or figure in the course depends on which edition you are reading.

The figures are built from the repository too. `scripts/build_course_map.py` draws the course map that opens every module, and `scripts/build_parameter_context.py` draws the source-data figures in Chapter 4 from the same Parquet files the Explorer runs on.

Chapter 1

How to use this course

In this module
You will see the deliverable the whole course builds toward, so you can judge every later module against it. You will calibrate your own starting point, in about ten minutes. Then you will pick the path that fits, so nobody sits through material they do not need.

Fast path

Short on time? Answer the three questions in Section 1.5 and read the path table in Section 1.6. That is about ten minutes, and it tells you which of the later modules you can skip.

1.1 The deliverable already exists, and a ministry has already published it

The document this course teaches you to write is an ordinary ministry-of-finance product. Many countries publish a fiscal risk statement each year, and at least one of them already carries a section built with this tool.

Uganda's Fiscal Risk Statement for FY 2024/25 is that section, and it is the example this course checks itself against. Pages 13 to 17 carry a chapter called "Climate Change Fiscal Risks." It reports a baseline fiscal path to 2099, five climate scenarios, and the GDP loss under each. It carries one headline finding: in the High, Hot and Vulnerable scenarios, public debt passes the 50 percent of GDP fiscal rule ceiling. The source line under the figures reads "QCRAFT (2023)."

That section came out of a five-day workshop. The IMF's Fiscal Affairs Department ran it for ministry staff in September 2023 and wrote it up as *Uganda: PFM Climate Assessment* (IMF, 2024). The tool that produced it is the tool this course teaches.

You will produce two things by the end of the course:

1. **An export packet.** The parameters you chose, a written rationale for each judgment call, and the polished charts and CSV that Q-CRAFT Explorer exports.

2. **A two-paragraph draft** in the register of your ministry's fiscal-risk documentation. In the worked example that document is Uganda's Fiscal Risks Statement. In your case it is whatever your ministry publishes, and if it publishes nothing yet, the Uganda section is the model.

That packet is the capstone, and it is what you defend to the senior officials who sign the document off. Chapter 7 contains the brief and the rubric it will be marked against. Every module in between either feeds that packet or tells you when not to trust it.

> **i** This course defers to the User Guide
>
> The IMF's User Guide (Tim and Rahman, 2024) remains the authoritative methodology reference. This course explains the concepts behind the tool and cites the User Guide section for each one, so you can go deeper where you need to.

1.2 By the end of this module you can

- **Locate** your own starting knowledge against three anchored descriptions of practice
- **Diagnose** which of three concept-inventory items you answered from intuition rather than from the debt identity
- **Select** the path (A, B or C) that matches your background, and justify the choice
- **Describe** the capstone deliverable in one sentence to a colleague who has not taken the course

1.3 Where you are in the course

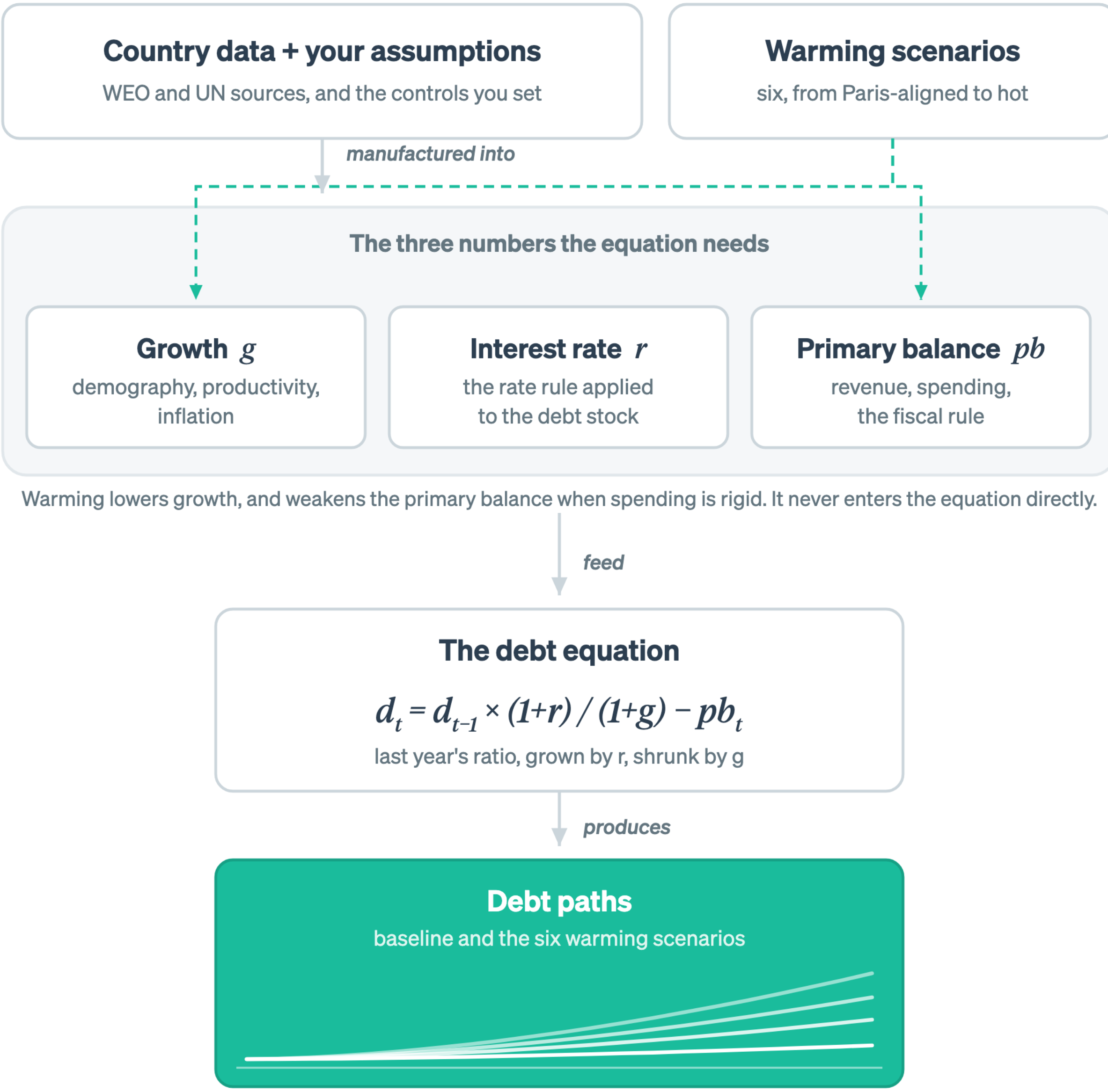


Figure 1.1: The course map. Country data and your assumptions are manufactured into the three numbers the debt equation needs. This module fixes the destination on the right.

Only the destination is lit, because that is the part to hold on to now. The same map opens every module, with that module's own nodes lit.

1.4 Self-assessment: which of these describes your Monday?

Three rows, one answer each: long-term fiscal projection, climate-fiscal analysis, and the tool itself. Pick the description closest to true. There is no scoring and no one sees it. The point is to route you, and to give you something to compare against in Chapter 7.

On long-term fiscal projection:

	Description
A1	I have not worked with a debt sustainability analysis. I know what the debt-to-GDP ratio is and roughly why it matters.
A2	I have read DSAs and can follow the tables. I could not build the projection myself without help.
A3	I have produced or reviewed a DSA for a country. I could set up a baseline projection this week if asked.

On climate-fiscal analysis:

	Description
B1	I work on climate policy or climate finance. I have not connected it to a debt projection.
B2	I have seen climate scenarios used in fiscal analysis and can describe what they are for.
B3	I have run or interpreted a climate-scenario fiscal projection for a country.

On the tool itself:

	Description
C1	I have never opened Q-CRAFT, in Excel or on the web.
C2	I have opened it and clicked around. I could not defend the parameter settings I used.
C3	I have run Q-CRAFT for a country and explained the output to someone else.

Write your three answers down. You will be asked for them again at the end.

1.5 Three questions before you start

Three questions follow, and they check which intuitions you are carrying in rather than what you have memorised. Answer each one, commit to it in writing, then open the answer.

! DRAFT FOR TEAL: concept-inventory question 1 (interest-growth differential)

Question. A country's government runs a primary balance of exactly zero: revenue equals non-interest spending, to the last unit of local currency. The nominal interest rate on its debt is 9 percent. Nominal GDP growth is 6 percent. Over the next year, the debt-to-GDP ratio will:

(a) Fall, because the government is not adding to its debt.
(b) Stay flat, because a zero primary balance is by definition debt-stabilizing.
(c) Rise, even though the government borrowed nothing new for programmes.

Answer

(c) Rise. Debt grows at the interest rate. The economy it is measured against grows more slowly. So the ratio climbs by roughly (9 - 6) / 1.06 in the first year, which is about 2.8 percent of the starting ratio. A country starting at 50 percent of GDP ends the year near 51.4 percent, having changed no policy at all.

If you picked (b), that is the most expensive misconception in this material: that a zero primary balance stabilizes debt. It stabilizes debt only when the interest rate equals growth. Every other time, the gap between the two is doing work someone has to pay for. Chapter 3 is built around that gap.

If you picked (a), you may be thinking of the deficit rather than the ratio. The stock of debt is flat in local currency. The ratio is not, because the denominator moved.

! DRAFT FOR TEAL: concept-inventory question 2 (expenditure rigidity)

Question. In Q-CRAFT Explorer you set expenditure rigidity to 1.0 and run a climate scenario. What have you assumed?

(a) Spending falls in step with GDP, holding the expenditure-to-GDP ratio at its baseline level.
(b) Spending stays at its baseline level in local currency even as GDP falls, so the whole revenue loss lands on the deficit.
(c) Spending is capped by the fiscal rule, whatever happens to GDP.

Answer

(b). Rigidity 1.0 means spending is sticky. The wage bill, pensions and existing commitments do not move when growth disappoints, so the whole climate-driven revenue shortfall becomes borrowing. That is the worst case, and the tool makes it the default on purpose.

If you picked (a), you have the scale inverted. That is rigidity 0.0. The inversion matters, because it flips the sign of your headline risk number. It is the easiest way to publish a wrong figure from this tool.

If you picked (c), you have merged two separate controls. The fiscal rule and expenditure rigidity are different parameters and they act on different scenarios. Chapter 4 separates them.

! DRAFT FOR TEAL: concept-inventory question 3 (what the output is)

Question. A Q-CRAFT run shows debt-to-GDP at 66 percent in 2099 under the Hot scenario against 47.5 percent in the baseline. A colleague asks what that means. Which answer is defensible?

(a) The country's debt will be about 66 percent of GDP in 2099 if the world warms on that path.
(b) Under a set of stated assumptions, climate damage of this magnitude would add roughly 19 percentage points of GDP to the debt ratio by 2099 relative to the same projection without climate damage.
(c) Climate change will cost the country 19 percent of GDP.

Answer

(b). The number that survives scrutiny is the *difference between two runs of the same model*, stated with its assumptions attached. Q-CRAFT compares scenarios. It does not forecast.

Answer (a) treats a 2099 projection as a prediction. Nobody forecasts 2099, and the tool's own User Guide says the results are not forecasts (pp. 5-6).

Answer (c) converts a debt-ratio gap into a cost figure, which it is not. The gap is the extra borrowing implied by lower growth under stated assumptions about how spending responds, and it leaves out disasters, sea-level rise and adaptation spending entirely. Chapter 6 is about that class of error.

The 47.5 and 66 percent figures come from the September 2023 IMF workshop with Uganda's ministry staff, reported in the C-PIMA high-level summary (IMF, 2024).

1.6 Pick your path

The three paths cover the same tool. They differ in how much of the economics is rebuilt for you along the way.

Path	Choose it if	Route	Rough time
A. Guided	You answered mostly A1/B1/C1. Common if you came to this from the climate side and not from the debt arithmetic.	M0 → M1 → **M2** → M3 → M4 → M5 → M6	5 to 6 hours
B. Standard	You answered mostly A2/B2/C1 or C2. Common if you know debt dynamics already and the tool is the new part.	M0 → M1 → M3 → M4 → M5 → M6	3 to 4 hours

Path	Choose it if	Route	Rough time
C. Fast	You answered A3 and C2 or C3. You have built projections and want the tool's specifics and the judgment calls.	M0 → M1 (early win only) → M3 self-checks → M4 → M5 → M6	2 hours

Chapter 3 exists for Path A. If you can already say what the interest-growth differential does to a debt ratio without looking it up, skip it. Forcing experienced staff through novice material makes the material harder, not easier.

Every module carries a fast-path marker near the top, telling you the shortest useful route through it.

> **A note on time**
>
> Budget three to four times whatever you think this will take. That ratio is the standard finding on expert time estimates for other people's learning, and it holds here. The ten-minute run in Chapter 2 does take ten minutes. The rest does not.

1.7 Wrapper: what you should have now

- The capstone in one sentence: an export packet with documented rationale, plus a two-paragraph fiscal risk draft.
- Three self-assessment answers written down for comparison in Chapter 7.
- Three concept-inventory answers, and a note on which ones you got from intuition rather than from the identity.
- A path.

On your desk this week: find the most recent fiscal risk documentation your own ministry publishes and read the section that would carry this analysis. If there is none, open Uganda's Fiscal Risk Statement FY 2024/25 at page 13. Either way, that is the genre you are writing into.

Chapter 2

What Q-CRAFT does and why it exists

In this module
You will run a full projection in ten minutes, before anything is explained. Then you will meet the single equation that produces the debt path to 2099, and the three numbers it needs: growth, the interest rate, and the primary balance. Everything else the tool does is manufacturing those three. The module closes with the parity evidence, so you can decide now whether to trust the numbers: baseline parity is exact for 147 of 147 tested countries.

Fast path

Short on time? Do Section 2.4 (the ten-minute run) and Section 2.6 (the three-number table and its self-check). Skip the rest and come back to it.

2.1 By the end of this module you can

- **Run** a full projection in Q-CRAFT Explorer for a country of your choice and export the results
- **Name** the three numbers the debt equation needs and **identify** what builds each one
- **Locate** any parameter you are asked about against the number it moves, without searching
- **Explain** to a colleague why a Python reimplementation of an Excel tool needs a parity test, and what "147 of 147" does and does not cover

2.2 Where you are in the course

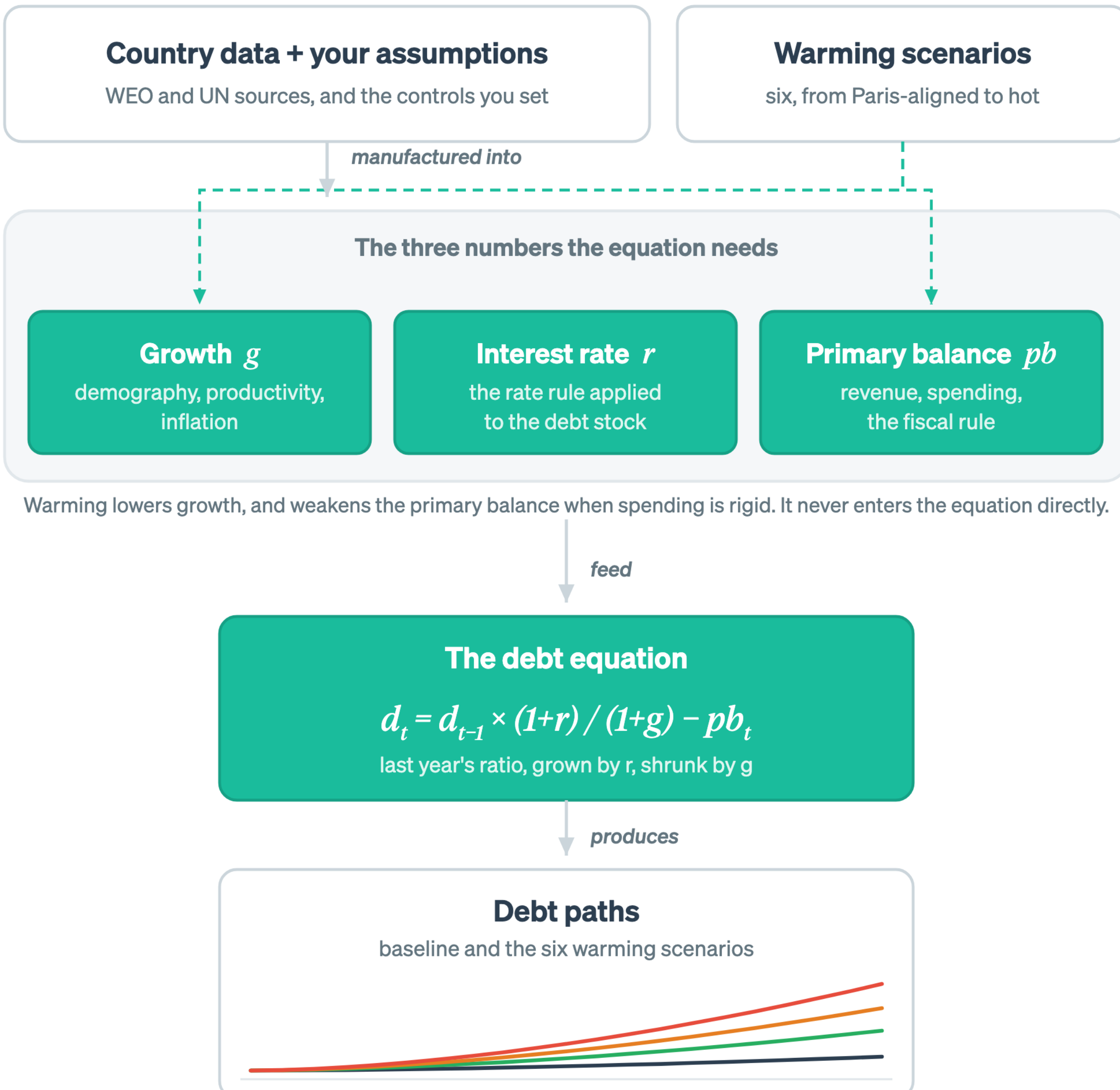


Figure 2.1: This module covers the middle of the chain: the three numbers, and the equation they feed.

Four nodes are lit: growth, the interest rate, the primary balance, and the equation that turns them into a debt path. Everything else in the course hangs off that group.

2.3 The Excel workbook calculates well and forgets everything

Building Q-CRAFT in Excel was the right call. Excel is everywhere. Every finance ministry has it, and putting it in front of someone costs no training budget. The workbook itself is well structured, clearly documented, and covers 197 economies. The team that built it made a tool that works.

The trouble starts after the file is saved. A fiscal economist in a low-income country's Ministry of Finance opens the workbook and finds 19 sheets. The macroeconomic data is pre-loaded from the IMF's World Economic Outlook, but the vintage may be a year old. The productivity growth assumption needs to reflect the country's structural outlook, and the workbook shows one default of 1.2 percent for every country. Was that default used in last year's C-PIMA assessment, or did the visiting IMF team change it? The workbook does not record that history.

A year later, the next analyst cannot tell a decision from a default. There is no audit trail, no version history, and no automated checks.

None of this is a failure of Excel. It is a limitation of a format built for calculation rather than for workflow, reproducibility, or analysis shared across institutions and years.

The cost lands in capacity development budgets. Countries need technical assistance to set up and interpret these projections, so the IMF sends economists to work alongside ministry staff. That work is valuable and expensive to scale. If practitioners could start from a higher base, with a tool that guides them through parameter selection and records their analytical choices, the same capacity development budget would go further.

Q-CRAFT Explorer puts the same economics on a platform that keeps the record. The calculation engine is a standalone Python package with extensive automated testing. The web interface loads bundled macroeconomic data automatically, currently the October 2024 WEO vintage, offers guidance at the point where you need it, and exposes the key projection parameters in a sidebar. The code is open source under the MIT license, so anyone can examine, verify, and extend it.

2.4 Run a country in ten minutes

Run the tool before you read how it works. The explanation lands better on something you have already seen move. Six steps follow, and the whole sequence takes about ten minutes.

> Zero to a projection
>
> **Open Q-CRAFT Explorer** and follow these steps:
>
> 1. **Select a country** from the sidebar dropdown. Pick the one you work on. If you want a run you can check against a published result, pick Uganda, which is the tool's verification country and the worked case in Chapter 5. Either way, WEO macroeconomic data and UN population projections load automatically. Nothing to download, nothing to paste.
> 2. **Read the sidebar context.** It shows the latest WEO debt-to-GDP ratio and total population. Check them against what you know before going further.
> 3. **Open the Baseline tab.** This is the country's fiscal trajectory with no climate damage. Note the debt-to-GDP number at 2099.
> 4. **Open the Climate tab.** Six warming scenarios, their effect on GDP. Note where the scenarios separate from each other, and in which year.
> 5. **Open the Analysis tab.** Baseline and all climate scenarios on one debt chart. The vertical gap between the baseline line and the Hot Unadapted line is the number your fiscal risk paragraph will be about.
> 6. **Open the Data tab and download both CSVs.** That is the first piece of your

capstone export packet. Keep it.

If you picked Uganda, you have now done with five clicks the analysis that ministry staff produced in a five-day workshop with a visiting IMF team in September 2023. The workshop's finding, published in the C-PIMA high-level summary: GDP loss by end of century could pass 4 percentage points, with debt rising to 66 percent of GDP against 47.5 percent in the baseline (IMF, 2024).

Your numbers will not match theirs exactly, even on the same country. That workshop used the 2023 WEO vintage and its own parameter choices, and the Explorer ships with WEO October 2024. Which vintage and which parameters were used is precisely the kind of thing that gets lost, and precisely what the export packet is for.

⚠ SCREENSHOT-TODO

Annotated screenshot of the Analysis tab for Uganda, the worked case, with the baseline-to-Hot-Unadapted gap called out on the chart. Do not ship this module without it: the sentence "the vertical gap is the number your paragraph is about" is doing visual work with no visual.

2.5 The equation needs three numbers, and the tool exists to build them

Q-CRAFT projects long-term fiscal outcomes under different climate scenarios, and the whole apparatus resolves to one equation with three inputs. The tool does not let fiscal policy feed back into GDP growth or interest rates: cut spending inside the model and growth does not respond. Economists call that a partial-equilibrium model (User Guide, p. 5).

2.5.1 The equation, in one sentence and then in symbols

Next year's debt ratio is this year's, grown by the interest rate, shrunk by economic growth, less whatever the government paid down.

That sentence is the whole model. The compact form is the sentence in shorthand:

$$d_t = d_{t-1} \times \frac{1+r_t}{1+g_t} - pb_t$$

Symbol	Meaning
d_t	Debt-to-GDP ratio in year t
r_t	Effective nominal interest rate on government debt
g_t	Nominal GDP growth rate
pb_t	Primary balance as a share of GDP (revenue minus non-interest spending)

Three of those four are inputs. The fourth, d, is the answer, and it is also last year's input, which is why the tool walks the projection forward one year at a time.

The sign of r minus g decides whether the ratio climbs on its own. When interest rates exceed growth, the debt ratio rises automatically, and the government must run a primary surplus even to hold debt still. When growth exceeds interest rates, the ratio declines even with modest deficits. Chapter 3 works this through with numbers. Skip it if that arithmetic is part of your working week.

Climate change reaches the equation through two indirect channels. It reduces GDP growth, and, when expenditure is rigid, it worsens the primary balance as revenue falls with GDP while spending stays elevated. There is no separate climate term. The climate impact sits inside g and pb, which is why finding it in the output takes a comparison rather than a single number.

Everything else the tool does is manufacturing g, r and pb.

2.6 Where the three numbers come from

Growth has three suppliers, the interest rate has one rule, and the primary balance has two sides. The table below is the thing to remember. The sections after it are for the number you happen to be arguing about.

The number	What builds it	What you control in V1
Growth, g	Working-age population, output per worker, and the price path, combined into nominal GDP growth	The UN population variant. Productivity and inflation sit at the Excel tool's defaults.
Interest rate, r	One of three rate rules, applied to the debt stock	Nothing. V1 uses the Excel default.
Primary balance, pb	Revenue, which tracks nominal GDP, minus primary expenditure, which grows with productivity, inflation and total population, plus any fiscal rule adjustment	The debt target, and the fiscal rule on or off
All three under climate	The same three, with damage applied to growth from 2030, and to the primary balance when spending is rigid	The climate scenario, and expenditure rigidity

Read the last column carefully. V1 exposes five controls, and the other assumptions sit at the Excel tool's defaults. That is a deliberate scoping choice rather than an oversight, and Chapter 6 says what it costs you.

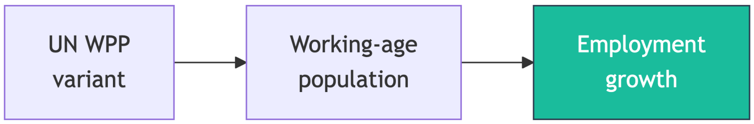
> ⚠ WIDGET-TODO: three-numbers-to-one-equation
>
> Embed the intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). The widget was specified as "seven-modules-to-one-equation" before the teaching frame changed. Rebuild it on the chain above, data and assumptions producing g, r and pb, with "seven modules" demoted to an architecture aside. Iframe under Quarto for now, native React once the site moves to Astro. The prose above must carry the full idea on its own, because the default state

of any interactive has to work for readers who never touch it.

2.6.1 Growth, g, is three things multiplied together

Growth is the number with the most suppliers, and the one your assumptions move most.

Demography sets how fast the workforce grows.



The workforce sets how fast the economy can grow. Q-CRAFT takes population by age group (children, working age, elderly) from UN World Population Prospects, in three variants: medium, high, and low. Growth in the working-age population, ages 15 to 64, drives employment growth after the WEO forecast horizon, and employment growth drives potential GDP. The contrast is sharp across the tool's country set. Ethiopia's working-age population is projected to keep growing for the rest of the century, which pushes potential growth up and debt ratios down. Thailand's has already turned and is falling, which does the opposite, and it does it before any climate damage is applied.

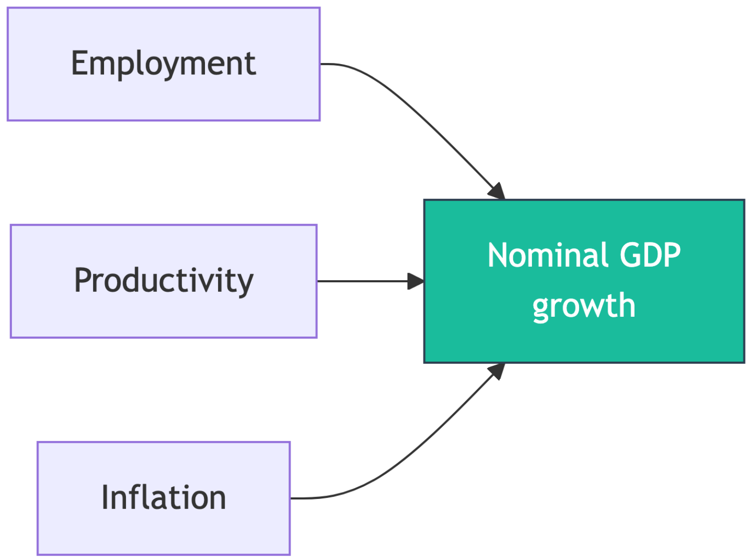
Productivity sets output per worker, and it moves gradually rather than in a step.

The measure is GDP per employed person, from World Bank data. You give the model a starting growth rate and a long-run rate, and it slides between them along an S-shaped path, which economists call a logistic curve. Vietnam is the case that makes the shape concrete: a catch-up economy raising output per worker quickly now, with the rate expected to ease as the level approaches the frontier. The resulting productivity level relative to the OECD is a useful realism check, because assumptions that imply a low-income country overtaking OECD productivity by 2099 deserve another look.

Inflation turns real growth into the nominal growth the debt ratio actually responds to.

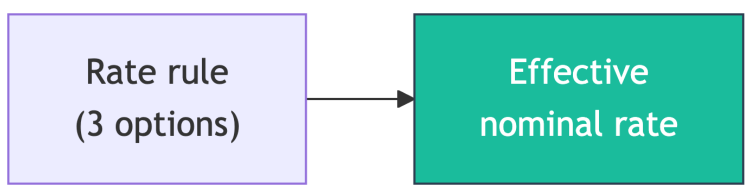
Inflation follows the same gradual S-shaped path as productivity. The starting rate should reflect the WEO projection for the final forecast year, and the ending rate should reflect the long-run target, typically 2 to 3 percent for advanced economies and 3 to 6 percent for emerging and developing economies. Where the central bank publishes an explicit target, that number is the reference point.

The three combine into nominal GDP growth.



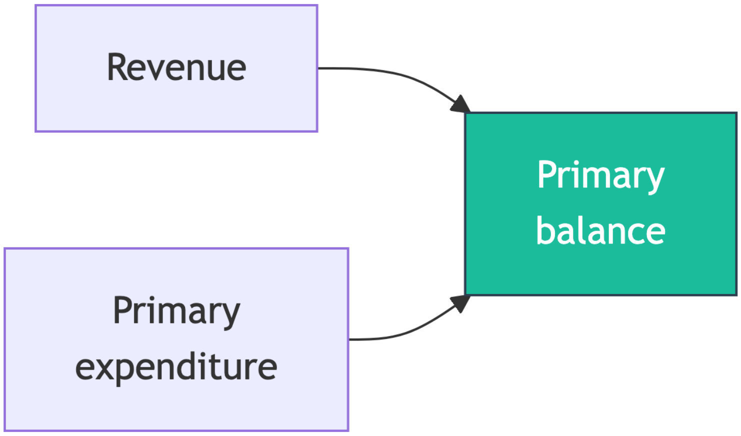
Nominal GDP growth decomposes into employment growth, productivity growth, and inflation. Through the WEO forecast period, which currently runs to 2029, Q-CRAFT uses the IMF's own projections. Beyond that horizon the model takes over, using the assumptions set in the dashboard.

2.6.2 The interest rate, r, is one rule applied to the debt stock



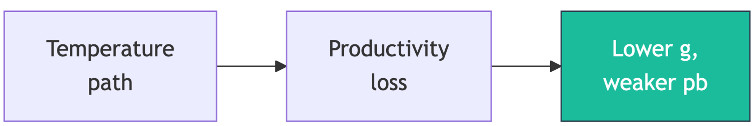
This number is what carrying the debt costs. Q-CRAFT offers three rules: a constant nominal rate, a constant gap between the interest rate and growth, or a constant real rate. Country expertise matters here, because the choice should reflect concessional lending terms, sovereign risk premia, and the maturity structure of domestic financial markets. For a country moving from concessional to commercial borrowing, this assumption does more work than any other. Zambia's shift into Eurobond issuance through the 2010s is the standard example, and the debt-service consequences arrived faster than any demographic or productivity trend could have.

2.6.3 The primary balance, pb, is revenue minus non-interest spending



The two sides grow on different rules. Revenue grows with nominal GDP, holding the revenue-to-GDP ratio constant. Primary expenditure, meaning spending excluding interest payments, grows multiplicatively with productivity, inflation, and total population: (1 + productivity growth) × (1 + inflation) × (1 + population growth), plus any fiscal rule adjustment. When the fiscal rule is enabled, it adjusts spending to steer debt toward a target ratio.

2.6.4 Climate scenarios move g, and pb when spending is rigid



Climate scenarios lower growth, and that is the only way climate enters the equation. The damage estimates are empirical, drawn from the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on the Kahn et al. (2021) methodology for long-term macroeconomic effects of climate change. Six scenarios run from Paris-aligned, below 2°C warming, to hot-unadapted, high warming with a slow policy response. Impacts begin in 2030, and before that year all scenarios match the baseline (User Guide, p. 19). The channel is productivity: reduced GDP growth means lower revenue, higher expenditure-to-GDP ratios, and accelerating debt.

The channel is temperature, so hot low-latitude economies take the largest modelled hit. Bangladesh shows both halves of that sentence at once. It is exactly the kind of economy the modelled channel bites hardest, and it is also the economy that shows what the channel leaves out, because sea-level rise is not in this model at all (Chapter 6).

ℹ The six climate scenarios

Scenario	Warming (wrt present)	Based on	Description
Paris-Aligned (1.5°C)	+0.7°C	SSP1-2.6	International commitments met, with significant emission cuts
Moderate (2°C)	+1.6°C	SSP2-4.5	Current trends continue, with no new aggressive mitigation
High (4°C+)	+2.5°C	SSP3-7.0	Countries scale back mitigation, response fragmented
Hot (3°C)	+3.5°C	SSP3-7.0 (90th pctile)	Same emissions as High, using the 90th percentile of temperature increases
Hot + Adapted	+3.5°C	SSP3-7.0 (90th pctile)	Same temperatures as Hot, with faster adaptation
Hot + Unadapted	+3.5°C	SSP3-7.0 (90th pctile)	Same temperatures as Hot, with very slow adaptation

Warming values are IPCC best estimates for 2081-2100 relative to present (User Guide Table 1). Adding about 1.1°C for present-to-pre-industrial warming gives the more familiar above-pre-industrial framing: Paris below 2°C, Moderate about 2.7°C, High about 3.6°C, Hot about 4.6°C.

One naming difference is worth knowing before a meeting. Uganda's Fiscal Risk Statement FY 2024/25 reports five of these and calls the last one "Vulnerable" rather than "Hot Unadapted". Same scenario, different label, and Chapter 5 uses the Explorer's naming.

These scenarios are conservative in ways that matter for how you write up the results. Chapter 6 sets out what they leave out.

2.6.5 The architecture, in one line

Under the hood the engine is seven pure functions, one per box in the sections above, composed into a single call. That is a fact about the code rather than about the economics, and it lives in the repository.

Self-check: where does it live?

A colleague sends you five questions in one email. For each, name which of the three numbers it moves, and where the tool builds that number. Commit to your five answers before opening the panel.

1. "Our fiscal rule ceiling is 50 percent of GDP. Does the model know that?"
2. "The projection assumes we keep growing at 6 percent forever. Does it?"
3. "Why does the debt path only start diverging across scenarios in 2030?"
4. "We are shifting from IDA credits to Eurobonds. Where does that show up?"
5. "Spending per person keeps rising in the projection. Who decided that?"

DRAFT FOR TEAL: self-check answers

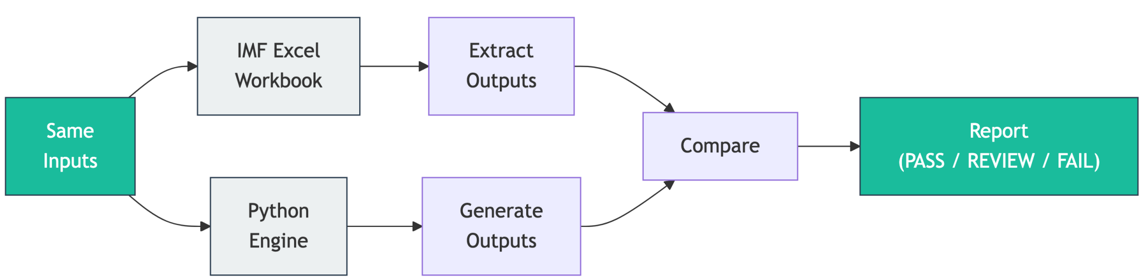
Four of the five land on two of the three numbers, which is the point of the exercise: most questions that arrive as five separate problems are two.

1. **The primary balance.** The debt target and the fiscal rule on/off switch both sit on the spending side, and both are yours to set.
2. **Growth.** It is built from employment, productivity and inflation along a gradual path, so it does not sit at the WEO rate forever.
3. **Growth, and the primary balance behind it.** The climate scenarios move both, and impacts start in 2030 by construction, with every scenario equal to the baseline before that (User Guide, p. 19).
4. **The interest rate.** The rate rule is where concessional-to-commercial transitions live. In V1 it sits at the Excel default, which is a limitation worth stating in your write-up.
5. **The primary balance.** Primary expenditure grows with productivity, inflation and total population, multiplicatively. Nobody decided it per person. It falls out of that formula.

If you missed 3, reread the climate section above. If you missed 5, the expenditure growth formula is the thing to hold: $(1 + a) \times (1 + b) \times (1 + c)$, never $a + b + c$.

2.7 The Python engine is checked against the Excel workbook, cell by cell

A reimplementation is worth nothing if it quietly disagrees with the original. This is the one place in the course where the verification story appears, and it is here because you should decide now whether to trust the numbers.



The verification pipeline: same inputs, two engines, automated comparison

Identical inputs go through both the original Excel workbook and the Python engine, and every output cell for every projection year is compared. Think of it as an auditor reconciling two copies of the same ledger. If a single number differs, the test fails and we investigate.

Where that stands. Baseline parity is exact for 147 of 147 tested countries: zero percentage-point deviation on debt-to-GDP, revenue, primary balance and primary expenditure as shares of GDP. An additional 25 parameter sensitivity combinations were tested and passed. Climate-scenario parity is confirmed for ratio metrics, meaning debt-to-GDP, revenue and expenditure as percent of GDP, across the tested countries and scenarios.

The second claim is narrower than the first, and the difference matters. Climate-scenario agreement has been established for the ratios you will quote in a Fiscal Risk Statement. It has not been established across every level series in the workbook. If your analysis turns on a climate-scenario figure in levels rather than as a share of GDP, check it against the Excel tool yourself.

The test suite is public and runs on every change. You do not have to take the claim on trust, because you can run it.

2.8 Wrapper: what you can now do

- You have run a full projection and exported the data. That file is capstone material.
- You can name the three numbers the equation needs and say what builds each one.
- You can say what the parity claim covers (baseline, 147 of 147, exact) and what it does not (climate levels).

Common error at this stage: treating the tool as seven things to learn. It is one equation with three inputs, and the inputs have suppliers. If you can answer the “where does it live” self-check, you have the map.

On your desk this week: when someone hands you a fiscal projection, ask which of r, g and pb their headline number moves. It works on projections that have nothing to do with climate.

Chapter 3

The debt equation

In this module
You will rebuild the debt identity from ordinary words, then check that the notation says the same thing. Two forces move the ratio: the gap between the interest rate and growth, and the primary balance. Everything Q-CRAFT does to model climate is an attack on one of those two, so this module is the foundation the rest of the course stands on.

Fast path

This is the Path A module. If you can already say what happens to a debt ratio when nominal growth falls below the effective interest rate, and why a zero primary balance does not stabilize debt, skip to Chapter 4. Nothing later depends on reading this.

3.1 Warm-up

Two questions from Chapter 2, from memory:

1. Which of the three numbers does demography move, and through what?
2. In which year do the climate scenarios start to differ from the baseline?

(Answers: g, through employment growth. 2030.)

3.2 The question from above is about trouble, not about 2099

The question that arrives from senior officials is rarely "what is the debt ratio in 2099." It is some version of: *if growth disappoints, how much trouble are we in, and how fast?*

That question has an exact answer, and it comes from one line of arithmetic that sits under every debt sustainability analysis. This module is that line, taken slowly.

3.3 By the end of this module you can

- **Derive** the debt-ratio identity in words, without notation, and check that the notation matches

- **Predict** the direction and rough size of the debt-ratio change given r, g and the primary balance
- **Compute** the debt-stabilizing primary balance for a country from three numbers
- **Explain** why climate damage shows up in a debt projection with no climate term in the equation

3.4 Where you are in the course

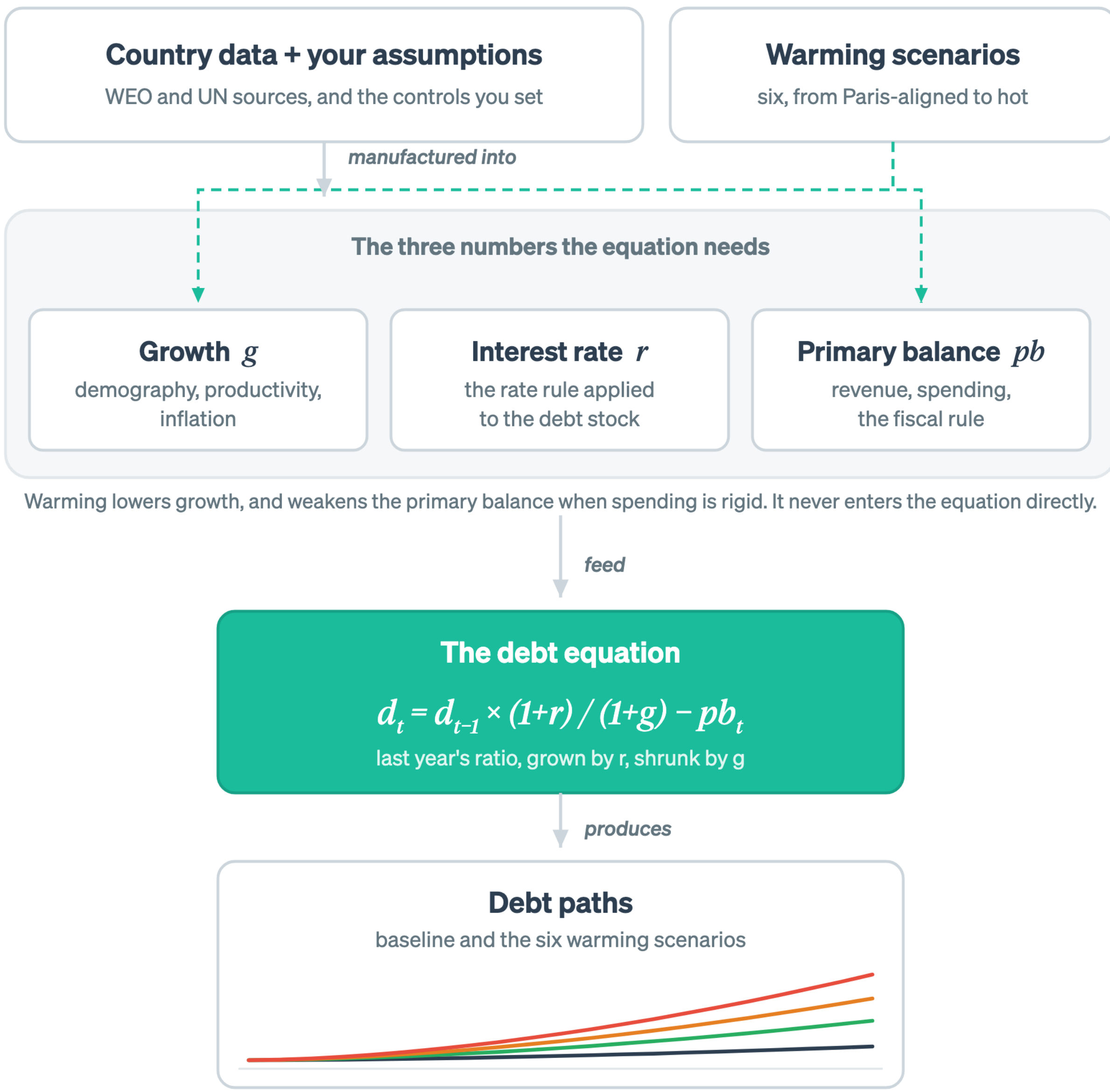


Figure 3.1: This module is the equation node, on its own.

One node is lit, and this module never leaves it.

3.5 Predict first

Three predictions before any arithmetic. Write your answers down, and check them at the end of the module.

1. A country's debt is 50 percent of GDP. It runs a primary balance of zero every year. Its effective interest rate is 9 percent, nominal growth 6 percent. Where is the ratio in ten years: below 50, near 50, or above 60?
2. Same country, but nominal growth is 11 percent. Same question.
3. Same country at 9 percent interest and 6 percent growth. What primary balance, as a share of GDP, would hold the ratio at exactly 50?

! DRAFT FOR TEAL: building the identity from words

Start with the stock, not the ratio. Government debt is a stock of local currency. Next year it is this year's stock, plus interest owed on it, plus any new borrowing to cover the gap between what the government spends on programmes and what it collects.

Debt next year = debt this year + interest on it + (non-interest spending − revenue)

That last bracket is the primary deficit. Flip its sign and it is the primary balance. Nothing controversial has happened yet.

Now divide by GDP, because that is how debt is judged. The numerator grows at the interest rate. The denominator grows at the rate of nominal GDP. Two things are racing, and the ratio moves according to which one is winning:

Debt ratio next year = debt ratio this year × (1 + interest rate) ÷ (1 + growth rate) − primary balance

Read the middle term as a scoreboard. If interest beats growth, the fraction is bigger than one and the ratio is amplified every year, whether or not anyone borrowed a unit of currency for a new road. If growth beats interest, the fraction is less than one and the ratio deflates on its own.

That is the equation. In shorthand:

$$d_t = d_{t-1} \times \frac{1 + r_t}{1 + g_t} - pb_t$$

The shape has a name, and you will recognise it elsewhere: a stock carried forward, amplified by a ratio of two competing forces, minus a flow. Compound interest with a headwind and a repayment.

The one sentence to keep. Next year's debt ratio is this year's, grown by interest, shrunk by growth, less what you paid down. If you can say that without looking, you own the equation.

3.6 The interest-growth gap moves the ratio with nobody borrowing

❗ DRAFT FOR TEAL: what the interest-growth differential does

The amplifier $\frac{1+r}{1+g}$ is usually reported as a single number, the interest-growth differential $(r-g)/(1+g)$. Its sign is the most consequential fact about a country's debt position, and in the short run it is not a policy choice.

Sign trace. Take a debt ratio of 50 percent of GDP and a primary balance of exactly zero, so the second term drops out and only the scoreboard is left.

r	g	Ratio after 1 year	After 10 years	What is happening
9%	6%	51.4	66.1	Interest wins. The ratio climbs with no new programme borrowing.
6%	6%	50.0	50.0	Dead heat. A zero primary balance stabilizes debt, and only here.
6%	9%	48.6	37.8	Growth wins. The ratio falls while the government runs a balanced primary account.

Three percentage points of differential, held for a decade, is the difference between 66.1 and 37.8 percent of GDP. Nothing in that table is a policy decision about spending. It is arithmetic on two macro variables.

> **WIDGET-TODO: interest-growth differential**
> Embed the interest-growth differential intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). Iframe under Quarto for now, native React once the site moves to Astro. The integration pass happens after this restructure and run 3 both land, and this callout is the anchor point. The prose above must carry the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

Why this is the climate channel. Q-CRAFT does not add a climate term to the equation. It lowers g. Look at what that does to the first row: every year of slower growth makes the amplifier larger, and the effect compounds. A tenth of a percentage point off growth, sustained for seventy years, is not a rounding error in this equation. It is the mechanism.

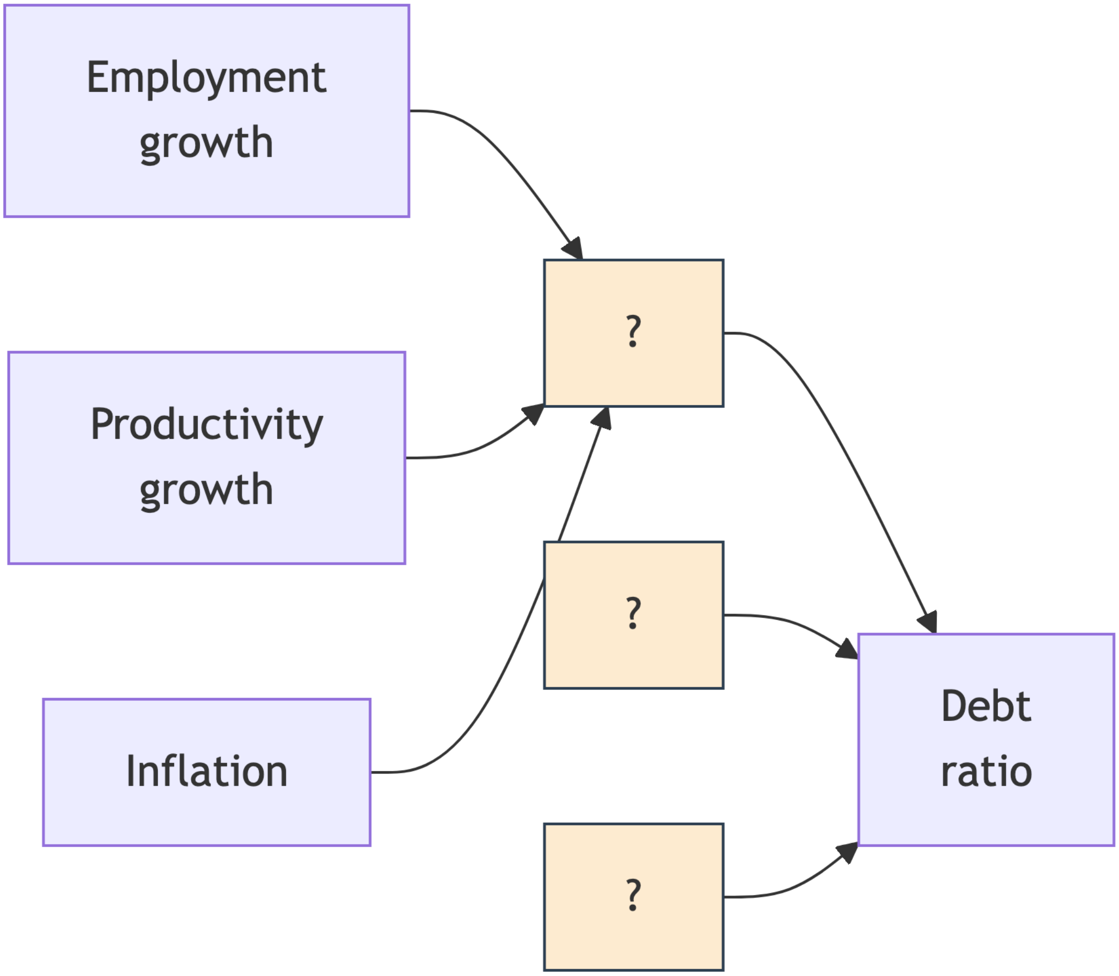
Sensitivity, ranked. If you have time to interrogate one assumption in a Q-CRAFT run, interrogate the interest rate rule, because it sets r and V1 does not expose it. If you have time for two, add the productivity growth path, because it sets most of g. Everything else moves the answer less.

3.7 The primary balance is the part the government controls

> **!** DRAFT FOR TEAL: the debt-stabilizing primary balance
>
> The second term is revenue minus non-interest spending, as a share of GDP. Interest is excluded because it is not discretionary this year. It is the price of last year's decisions.
>
> **The question the term answers.** Given where the scoreboard sits, what primary balance holds the ratio still? Set next year's ratio equal to this year's and solve:
>
> $$pb^* = d \times \frac{r - g}{1 + g}$$
>
> That is the debt-stabilizing primary balance, a standard DSA quantity. It says that the worse your interest-growth differential and the higher your existing debt, the larger the surplus you need to stand still.
>
> **Run it for the country in the table.** Debt at 50 percent of GDP, r at 9 percent, g at 6 percent:
>
> $$pb^* = 0.50 \times \frac{0.09 - 0.06}{1.06} = 0.0142$$
>
> The arithmetic asks for a primary surplus of about 1.4 percent of GDP, every year, to keep the ratio at 50. A government running a primary deficit of 1 percent in that position is 2.4 percentage points of GDP away from stable.
>
> **Perspective.** Put it against revenue rather than GDP, because that is the pot it comes out of. A country collecting 14 percent of GDP in revenue has to find that 1.4 percent from the same envelope, so a tenth of everything it raises goes to standing still. That is what the arithmetic asks for before a single new priority is funded.
>
> **Now connect it to rigidity.** Climate damage lowers g, which raises pb^*, so you need a bigger surplus. It also lowers revenue, because revenue tracks nominal GDP. If spending does not fall to match, the actual primary balance moves the wrong way at the same time as the required one moves up. Those two effects push in the same direction, which is why the expenditure rigidity parameter in Chapter 4 has such a large effect on the answer.

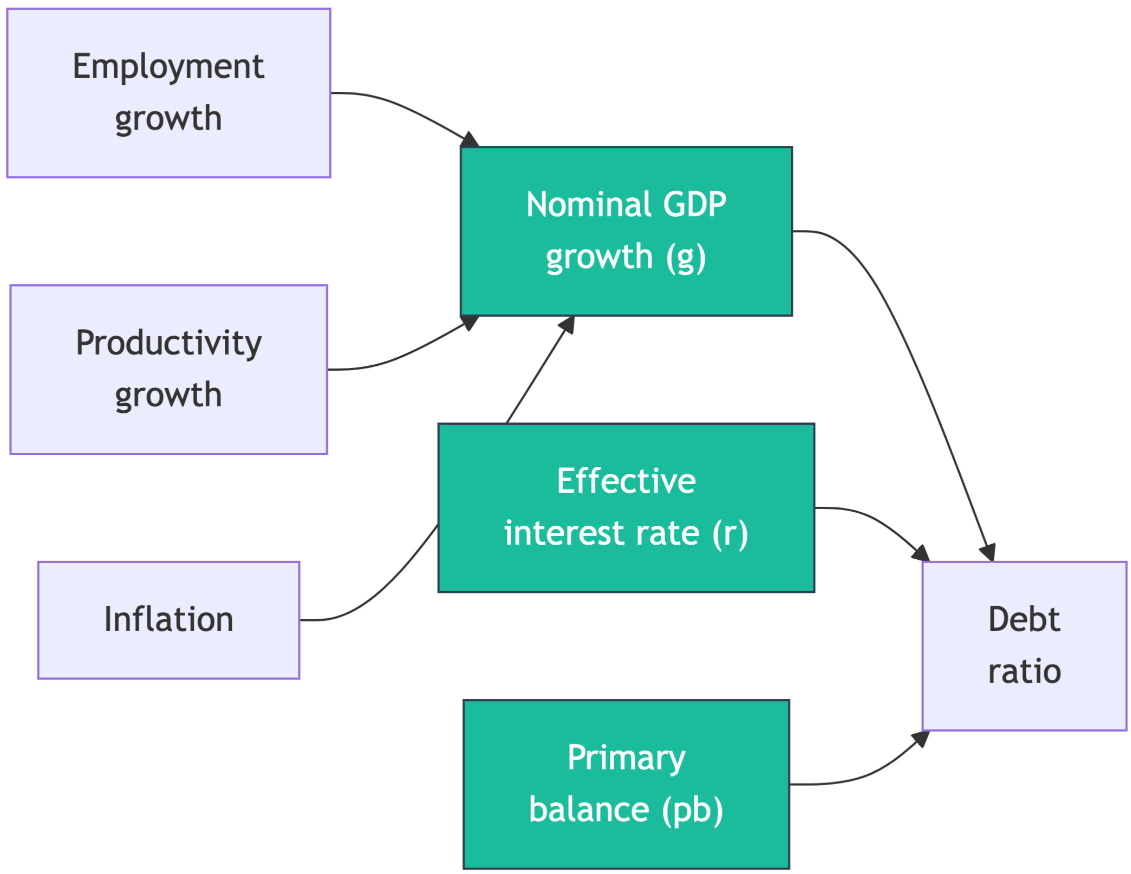
3.8 Complete the map

The debt equation appears below as a diagram with three nodes left blank. Fill them in before opening the answer, because constructing the map is the exercise and reading a finished one is not.



Fill in the three blanks.

The completed map



Three inputs, one output. If your version had climate as a fourth arrow into the debt ratio, redraw it: climate enters through g and through pb, never directly. That is the single most useful structural fact in this course, and it is why reading a climate result takes a comparison between two runs.

3.9 Predict, then run it

Now use the app, and predict before each click. The prediction is what makes the observation stick.

Predict, observe, explain

Setup. Open Q-CRAFT Explorer, select a country you know well, fiscal rule **off**.
Predict. Write down: over 2030 to 2099, is that country's baseline debt ratio rising, flat, or falling? One sentence on why, naming r and g.
Observe. Open the Baseline tab. Read the debt-to-GDP trajectory. Then look at the Fiscal Balances chart and note whether the primary balance is in surplus or deficit.
Explain. Reconcile the two. If the primary balance is in deficit and the ratio is not exploding, the scoreboard is working in that country's favour: nominal growth is beating the effective interest rate, which is the normal position for a country still borrowing largely on concessional

terms. If your prediction assumed a deficit must mean rising debt, that is the misconception from Chapter 1 question 1, and now you have seen it fail on real data.

Then run a second country that borrows commercially, and watch the same chart tell the opposite story. Uganda and Zambia make the contrast quickly. The arithmetic did not change between them, and the sign of r minus g did.

⚠ SCREENSHOT-TODO

Baseline tab with the debt trajectory and the Fiscal Balances panel side by side, for a country where the primary balance is in deficit and the ratio is stable, so the reconciliation is visible in one image. Uganda works.

3.10 Self-check: three judgment calls

A colleague makes three claims

Your counterpart in another department sends these. For each, say whether you would agree, and what you would check.

1. "Our debt ratio fell last year, so the fiscal position improved."
2. "We should compare our primary deficit to the regional average to see if we are overspending."
3. "Inflation is helping us: it inflates away the debt, so higher inflation is good for the ratio."

   DRAFT FOR TEAL: self-check answers

   All three are the same mistake in different clothes: reading one number without the arithmetic that gives it meaning.

   1. **Not necessarily.** A ratio can fall on the denominator. Uganda's own Fiscal Risk Statement makes this point about FY2022/23: the debt ratio fell to 46.9 percent from 48.4, and the statement attributes the fall partly to nominal GDP rising on high inflation, warning that the effect would subside as inflation fell. Check whether the stock of debt fell or the denominator moved.
   2. **Wrong benchmark.** The primary deficit that is sustainable depends on your own r, g and starting debt, not on what neighbours do. The comparison that means something is your actual primary balance against your own debt-stabilizing primary balance, $d \times (r - g)/(1 + g)$. Two countries with identical deficits can be in opposite positions.
   3. **Half right, and the half that is wrong is expensive.** Unexpected inflation does raise nominal g and deflate the ratio, once. But it feeds into the effective interest rate on new and floating-rate borrowing, so r follows g up with a lag and the gain reverses. In Q-CRAFT specifically, revenue and primary expenditure both grow with inflation, so a permanently higher inflation assumption largely cancels out of the ratio. If you want to see this, it is a good use of the sensitivity runs in Chapter 4.

3.11 Check your predictions

Back to the three predictions from the start of the module.

1. **Above 60.** 66.1 percent after ten years. Zero primary balance is not stabilizing when $r > g$.
2. **Below 50.** 37.8 percent. The same zero primary balance, the opposite result, because the scoreboard flipped.
3. **A surplus of about 1.4 percent of GDP**, from $0.50 \times (0.09 - 0.06)/1.06$.

If you got all three, you did not need this module and Chapter 1 routed you wrong. If you got the third one only after seeing the formula, that is the expected outcome and you are ready for Chapter 4.

3.12 Wrapper: what you can now do

- You can build the debt identity from a sentence and check that the notation matches.
- You can compute a debt-stabilizing primary balance from debt, r and g.
- You can say where climate enters the equation, which is nowhere directly and everywhere through g and pb.

Common errors on this material: reading a falling ratio as fiscal improvement, assuming a balanced primary account stabilizes debt, and drawing climate as a separate term.

One thing this module deliberately skipped: the baseline applies a floor of zero to the debt ratio and the climate scenarios do not. That asymmetry changes how you read the bottom of a fan chart, and it is covered in Chapter 6.

On your desk this week: compute the debt-stabilizing primary balance for your own country from the current WEO numbers, and compare it to the actual. It is three numbers and one division, and it is the sentence that opens most credible fiscal risk write-ups.

Chapter 4

Choosing the parameters

In this module
You will set all five of the Explorer's controls for a country and write down why you set each one where you did. Every parameter gets the same treatment: what it is, why it matters, how to set it, what the underlying source actually shows, a predict-observe-explain run in the app, and one judgment call to defend. The rationale lines you write here become the assumption section of your export packet, and they are what survives you leaving the ministry.

Fast path

Already run the tool? Do the five **Defend your choice** self-checks and the five **Document it** blocks, and skip the What/Why prose. That is about 25 minutes and it is the part the capstone rubric marks.

4.1 Warm-up

From Chapter 2, without looking:

1. Which of the three numbers do the debt target and the fiscal rule move?
2. Expenditure rigidity of 1.0 means spending is what?

(Answers: the primary balance, through the spending side. Sticky, so the full revenue loss lands on the deficit.)

4.2 The assumption nobody wrote down

Reopen the vignette from Chapter 2. A year after the projection was built, nobody can say whether 1.2 percent productivity growth was a considered view about the country or a default nobody touched. The analysis is not wrong. It is unusable, because it cannot be defended.

Q-CRAFT Explorer cannot stop you leaving a default in place. What it can do is make the record, so your parameters and your reasons travel with the output. This module is about earning the right to that record.

4.3 By the end of this module you can

- **Set** all five Explorer parameters for a country and **justify** each against country evidence
- **Predict** the direction and rough magnitude of each parameter's effect before you move the slider
- **Defend** an expenditure rigidity choice against a colleague who prefers a different one
- **Produce** the assumption-rationale section of the export packet, one entry per parameter

4.4 Where you are in the course

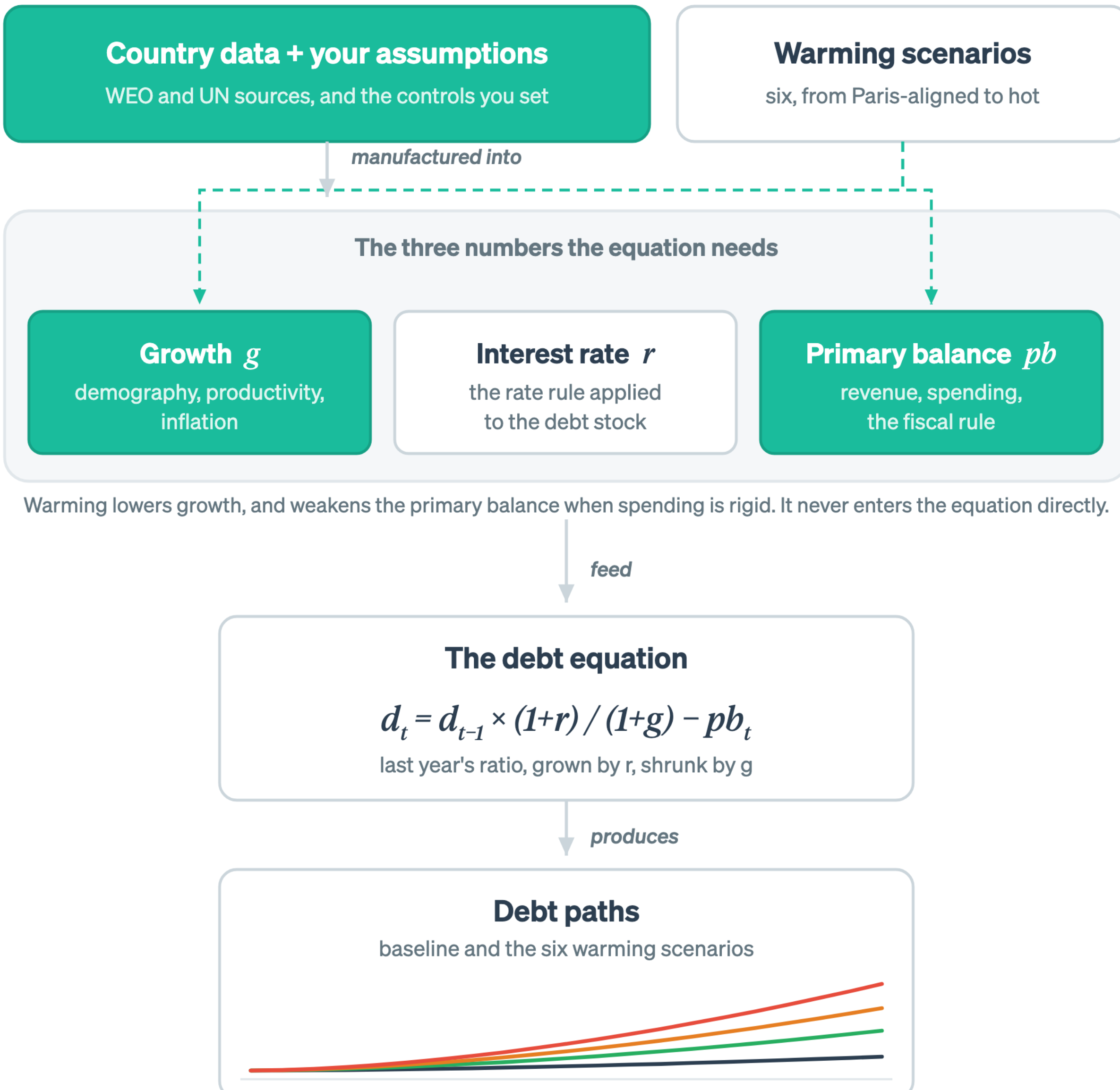


Figure 4.1: This module is the start of the chain: the data you load, the assumptions you set on top of it, and the two numbers they move.

Three nodes are lit. The data you load and the controls you set on top of it move growth and the primary balance. The interest rate is dark because V1 does not expose it.

4.5 Four tabs, and what each one is for

Four tabs hold everything the tool shows you, and you will be moving between them and the sidebar constantly. The **Baseline** tab is the no-climate-change projection. The **Climate** tab shows what warming does to GDP, scenario by scenario. The **Analysis** tab overlays baseline and climate debt paths on one chart. The **Data** tab gives you the grid and the CSV exports. Reading those charts properly is Chapter 5, and for now you only need to know where to look when a parameter moves.

4.6 Five controls, and four of them shape the projection

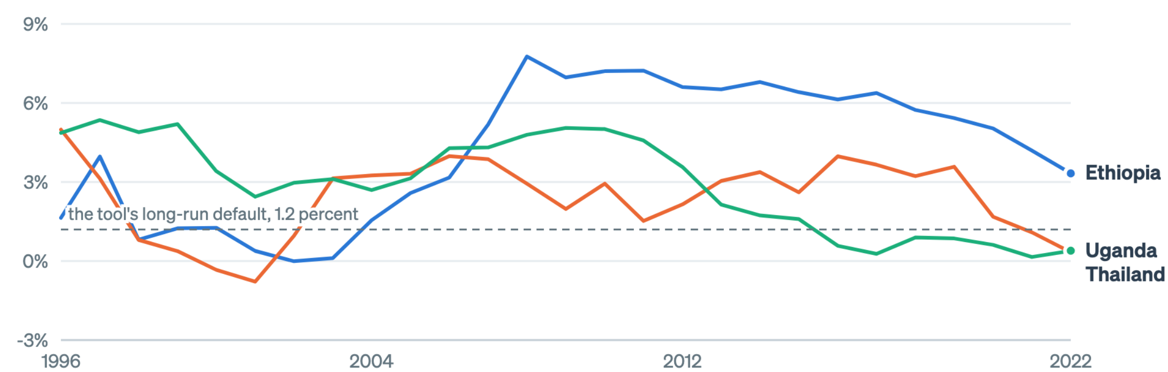
Q-CRAFT Explorer has five user-facing parameters. The first, country selection, loads the data. The remaining four shape the projection.

Productivity, inflation and interest rate assumptions are not exposed in V1, because they sit at defaults matched to the original Excel tool. That is a limitation with consequences, and Chapter 6 says what they are. It also means the four parameters below carry more of your analytical judgment than they would in the full workbook.

Two of those three defaults are worth seeing against the record they are meant to continue. Productivity growth slides from 5.0 percent to 1.2 percent, and inflation from 5.0 percent to 3.5 percent, for every country in the tool.

What output per worker has actually done

Growth in GDP per employed person, five-year trailing average. V1 slides every country from 5.0 percent to 1.2 percent.



Source: World Bank World Development Indicators, as bundled with Q-CRAFT Explorer.

Figure 4.2: V1 slides productivity growth from 5.0 percent to 1.2 percent for every country. Set that path against what output per worker has actually done before you quote a result that depends on it.

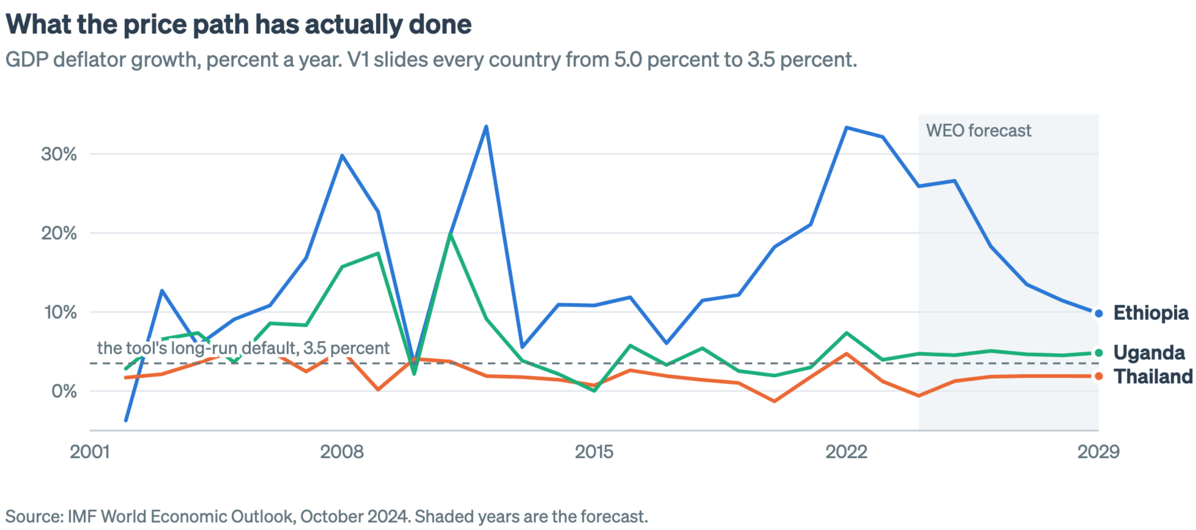


Figure 4.3: V1 slides inflation from 5.0 percent to 3.5 percent for every country. The three records here start in very different places, which is what the single default is averaging over.

Neither figure tells you the default is wrong. Both tell you what it is averaging over, which is the thing to say out loud when a result turns on either assumption. Where the gap between your country's record and the default is large, the sentence for your write-up is that V1 uses the Excel tool's global default and your country sits somewhere else, with the direction named.

> **What the source shows, and where to find more of it**
>
> Every figure in this module is drawn from the same Parquet inputs the Explorer runs on, by `scripts/build_parameter_context.py` in the repository. The Explorer is gaining interactive context panels that do the same job at the point of decision, so you can look at the source data for your own country while you set the control rather than afterwards.

4.6.1 Country selection

What it is. The country dropdown lists every country for which Q-CRAFT Explorer has complete data coverage across all required input datasets. Selecting a country loads macroeconomic data from the IMF's World Economic Outlook, currently the October 2024 vintage, and population projections from the UN World Population Prospects, 2022 revision.

Why it matters. The country you select determines all historical data and every WEO forecast-period projection. There is no manual data entry: the tool loads real GDP, nominal GDP, revenue, expenditure, debt, interest rates, and demographic projections automatically.

How to set it. Choose the country you are analyzing. The sidebar then displays two pieces of context: the latest WEO debt-to-GDP ratio and total population. Use them as a sanity check, and investigate before proceeding if either number does not match your understanding.

Country coverage

Q-CRAFT Explorer currently covers 197 countries, and coverage is expanding as verification progresses. If your country is not listed, it may be added in a future update. See the IMF User Guide (Tim and Rahman, 2024), Section II.B for details on what data is loaded for each country.

What the source shows. The starting debt ratio is not a small detail, because the whole projection compounds off it.

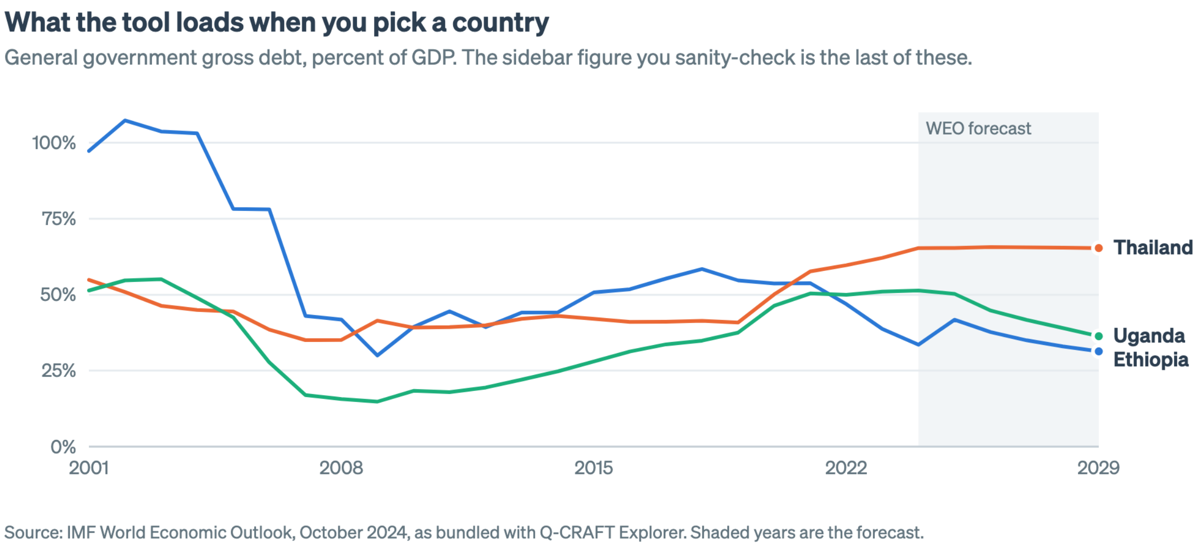


Figure 4.4: Three countries the tool already holds data for. Debt ratios differ by more than a factor of three across them, which is why the first thing you do after selecting a country is check the number the sidebar reports.

Predict, observe, explain

Predict. Before you select your country, write down the debt-to-GDP ratio you expect the sidebar to report, from your own knowledge of its latest position.
Observe. Select the country and read the sidebar figure.
Explain. If your number and the tool's differ by more than a point or two, work out which one is on a different basis before you go any further. Three causes are common: gross versus net debt, fiscal year versus calendar year, and the WEO vintage lagging the ministry's own outturn. Each one is a legitimate difference, and each one will be the first question you are asked about your results.

Defend your choice: country and vintage

A reviewer asks: "Is this the current debt number?" Take the worked case. You are using WEO October 2024, and Uganda's own FY2024/25 Fiscal Risk Statement reports public debt at 46.9 percent of GDP at June 2023 and projects 49.2 percent at June 2024. What do you say?

DRAFT FOR TEAL: a defensible answer on vintage

Name the vintage, name the basis, and state which direction the difference runs. "This is WEO October 2024 on a calendar-year basis. The ministry's June 2024 figure is 49.2 percent on a fiscal-year basis. The starting point differs by under a point, and the climate result is a difference between two runs from the same starting point, so it is not sensitive to that gap." The last clause is the one that ends the conversation, and it is only available to you because you understand what the tool computes.

Document it

In your export packet, record: **data vintage** (WEO October 2024, UN WPP 2022), **starting debt ratio as loaded**, and **any known difference from your ministry's own figure, with the reason**.

4.6.2 Demography variant

What it is. The UN publishes population projections in three variants, Medium, High and Low, reflecting different fertility assumptions. This parameter selects which variant Q-CRAFT uses for population and labor force projections through 2100.

Why it matters. Working-age population growth, ages 15 to 64, drives employment growth in Q-CRAFT's production function after the WEO forecast horizon. Higher population growth means a larger labor force, higher potential GDP, and, all else equal, more favourable debt dynamics. Countries facing demographic decline see slower growth and harder fiscal trajectories. Total population growth matters separately, because primary spending grows with it (see Chapter 2 for the expenditure growth formula).

How to set it. Three variants, and one of them is the default choice:

- **Medium** is the standard assumption for most policy analysis. Use it unless you have a specific reason to explore demographic scenarios.
- **High** and **Low** bracket the uncertainty. They earn their place for countries where fertility trends are uncertain or where demographic policy changes are anticipated.
- For countries in rapid demographic transition, with a declining working-age population, compare Medium against Low to see how sensitive the fiscal path is to the demographic outcome.

For detailed methodology, see the IMF User Guide, pp. 10-12 and Section IV.A on demography and employment.

What the source shows. The three variants are the UN's own account of how uncertain its projection is, and how far apart they run depends on where the country is in its demographic transition.

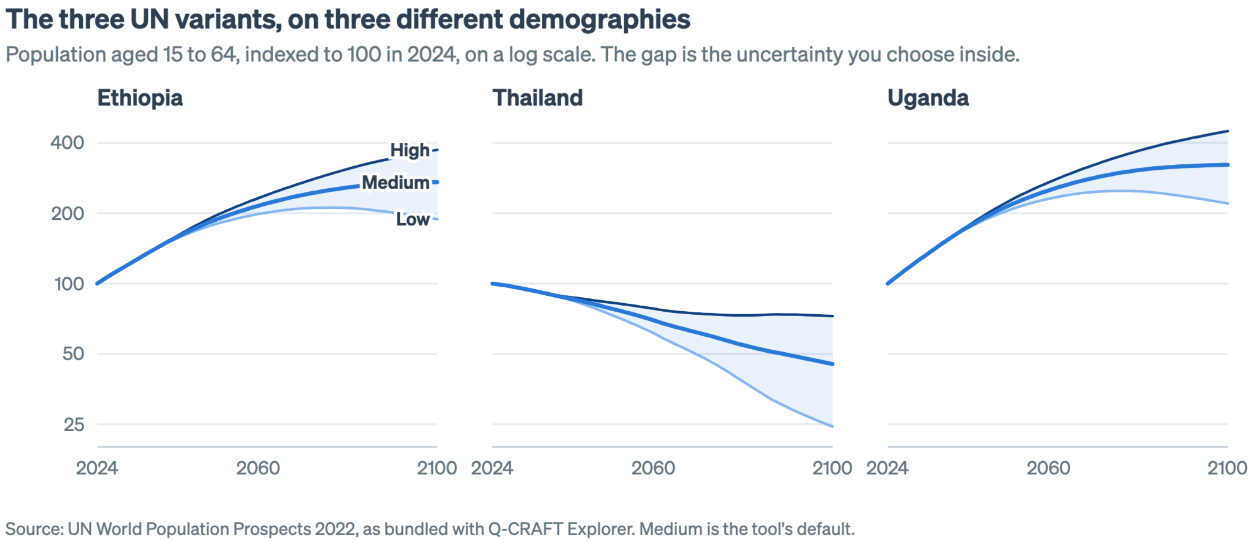


Figure 4.5: The demography control picks one of these three lines. In a country whose working-age population is still climbing the variants separate slowly. In one where it has turned down, the choice between Low and High is the difference between a workforce at a quarter of today's and one at three quarters.

Predict, observe, explain

Predict. Ethiopia has one of the youngest populations in the world, and Thailand's working-age population has already started to fall. Before you touch the control, will switching from Medium to High move Ethiopia's 2099 debt ratio a lot, a little, or barely at all? Write down a direction and a rough size in percentage points, then write a second prediction for Thailand.
Observe. Run Ethiopia, switch between Medium and High, and watch the Baseline tab's debt trajectory. Then run Thailand and do the same. Compare the size of the two moves.
Explain. In countries with young, growing populations the variants barely separate over the projection, because the working-age share is already rising in every one of them. In countries facing demographic decline the gap widens sharply: slower working-age growth means lower GDP growth and rising expenditure-to-GDP ratios at the same time. Both effects run through g. The expenditure channel runs through total population and the growth channel runs through working-age population, and those two populations move differently, which is why the net effect is not obvious in advance.

Defend your choice: demography

A colleague argues you should use the Low variant, because your country's fertility rate has been falling faster than the UN projected. Do you switch?

> **DRAFT FOR TEAL: a defensible answer on the demography variant**
> Probably not for the headline run, and certainly worth a sensitivity run. Medium is the reference variant, and using anything else as your headline puts you in the position of defending a demographic forecast inside a fiscal risk document. That is not the argument you want to be having. The colleague's point is testable in

two minutes: run Low, record the difference in the 2099 debt ratio, and put the number in the packet as a sensitivity. If the difference is small, you have retired the objection with evidence. If it is large, you have found something worth a paragraph.

Document it

Record: **variant chosen**, **why** (one line), and **the 2099 debt ratio under at least one alternative variant** as a sensitivity.

4.6.3 Debt target (% of GDP)

What it is. The debt target is the debt-to-GDP ratio the fiscal rule steers toward over time. The range runs from 0 to 200 percent of GDP.

Why it matters. The target does nothing until the fiscal rule is enabled, and then it determines how hard spending has to adjust. A target below current debt calls for consolidation, which cuts primary expenditure. A higher target calls for less adjustment.

How to set it. Start from the country's actual fiscal framework. Many countries have explicit fiscal rules with legislated debt targets, and you should use those where they exist. Uganda's FY2024/25 Fiscal Risk Statement reads its climate results against a 50 percent of GDP fiscal rule ceiling, and reports that public debt passes it in the High, Hot and Vulnerable scenarios. Where no formal target exists, the debt-stabilizing primary balance is a useful benchmark, and it is a standard DSA metric (see the IMF User Guide, Section IV.A, and Chapter 3).

Three rough starting points, none of them authoritative IMF guidance:

- **Low-income countries (LICs):** 40 to 50 percent of GDP is a common range in practice, broadly consistent with IMF-World Bank Debt Sustainability Framework thresholds.
- **Emerging markets (EMs):** 50 to 70 percent of GDP, depending on the country's market access, institutional quality, and fiscal track record.
- **Advanced economies (AEs):** can sustain higher ratios, 60 to 100 percent and above, though the appropriate target depends on growth prospects and market confidence.

For detailed methodology, see the IMF User Guide, pp. 15-18 on fiscal rule assumptions and the baseline scenario.

What the source shows. Nothing, in this case, and that is worth being clear about. The target is a policy choice rather than a published series, so there is no source figure for it. The nearest thing to evidence is the country's own record, which is the debt-ratio figure at the top of this module, read against whatever anchor your fiscal framework already sets.

Predict, observe, explain

Predict. With the fiscal rule enabled, you are about to move the debt target from 40 percent to 80 percent. Write down what happens to primary expenditure in the early years, and what happens to the 2099 debt ratio.
Observe. Set the target to 40, read the primary expenditure path and the debt path in the Baseline tab. Set it to 80 and read them again.

Explain. A lower target forces faster consolidation, so expenditure is cut harder to bring debt down. A higher target leaves more spending room. The adjustment is gradual rather than immediate, because it works through the fiscal gap over several years. If you predicted an instant jump in expenditure, correct that: the rule is a feedback loop keyed to last year's debt position, not a switch.

! Defend your choice: the target

You are asked to run the projection against a 40 percent target rather than the 50 percent ceiling your fiscal risk documentation already uses, "to be prudent." What do you do, and what do you say?

> **DRAFT FOR TEAL: a defensible answer on the debt target**
> Run both, and be explicit about what the target is doing in the model. In Q-CRAFT the debt target is not a judgment about what is sustainable. It is the anchor the simulated fiscal rule steers toward, so changing it changes the assumed policy reaction rather than the country's risk. The honest framing gives both numbers: "At the 50 percent anchor, the Hot scenario breaches the ceiling by X. At a 40 percent anchor, staying inside it requires Y percent of GDP a year of additional primary balance." That answers the request without smuggling a policy preference into a risk assessment.

i Document it

Record: **target used**, **its source** (legislated rule, Charter for Fiscal Responsibility, DSF threshold, analyst judgment), and **the alternative target you tested**.

4.6.4 Fiscal rule (Yes / No)

What it is. The fiscal rule, when enabled, adjusts primary expenditure to steer debt toward the target ratio. When it is disabled, spending follows baseline trends whatever the debt level does.

Why it matters. Without a fiscal rule, debt dynamics are purely mechanical: they follow the interest-growth differential and whatever primary balance falls out of revenue and expenditure trends. With the rule on, the model simulates active fiscal policy, so the government cuts spending as debt rises or spends more as fiscal space opens.

How to set it. Run it both ways, in this order:

- **Start with the fiscal rule off** to see the "no policy change" baseline. That reveals the underlying trajectory: is debt stable, rising, or falling under current trends?
- **Turn the fiscal rule on** to see how active policy changes the picture. That answers a different question: how much consolidation would reaching the debt target take?
- Most countries have some form of fiscal anchor in practice, though enforcement varies. The IMF User Guide notes that Q-CRAFT's fiscal rule is an approximation, because it does not model the GDP effects of fiscal consolidation (User Guide, p. 17, footnote 11).

The adjustment lands on the level of primary expenditure rather than on its growth rate. It is applied after the multiplicative growth factors, and it depends on the prior year's state, which is why Q-CRAFT computes this recursively year by year.

Predict, observe, explain

Predict. Run your country with the rule off first. Predict whether turning the rule on will change the 2099 debt ratio a lot or a little, and say why in terms of where the off-rule path ends up relative to the target.
Observe. Run the same country with the fiscal rule on and off. Watch both the debt path and the primary expenditure path.
Explain. With the rule off and unfavourable debt dynamics, meaning the interest rate exceeds growth, debt-to-GDP can rise without bound. Turn the rule on and debt converges toward your target, with the primary expenditure chart showing the spending cuts required to get there. The gap between the two runs is the fiscal effort your target implies. If the off-rule path already sits near the target, turning the rule on does almost nothing, and that is itself a finding worth reporting.

! Defend your choice: rule on or off

Which run belongs in a Fiscal Risk Statement: rule on or rule off?

DRAFT FOR TEAL: a defensible answer on rule on or off

Both belong, because they answer different questions. Rule **off** is the risk statement: this is where we end up if policy does not respond, which is the honest baseline for a document whose job is to disclose risk. Rule **on** is the policy statement: this is the effort required to hold the anchor. Publishing only the rule-on run understates the risk, because it assumes the response you are trying to justify. Publishing only the rule-off run invites the objection that no government would sit still for seventy years. Report the pair and say which is which.

Document it

Record: **rule on or off for the headline run**, **the paired run**, and **one sentence on which question each answers**.

4.6.5 Expenditure rigidity (0.0 - 1.0)

What it is. Expenditure rigidity measures how far government spending resists downward adjustment when climate shocks reduce GDP. It affects climate scenarios only, never the baseline.

Why it matters. This is the parameter that decides how much of the climate cost becomes debt. When climate change reduces GDP, government revenue falls with it, because revenue tracks nominal GDP. What happens to spending then determines how much of the shortfall shows up as additional borrowing.

- **1.0 (fully rigid, worst case):** spending stays at its baseline level in local currency terms even as GDP falls. Revenue declines and expenditure does not adjust, so the entire climate impact passes through to the primary balance and onto the debt ratio. This is a government that cannot or will not cut spending when growth disappoints, because of a large civil service wage bill, entitlement programmes, or political constraints on adjustment.

- **0.0 (fully flexible):** spending falls in proportion with GDP, holding the expenditure-to-GDP ratio at its baseline level. The debt impact of climate change is smaller, because the government absorbs the shock by reducing real spending per person.
- **Values between 0 and 1** represent partial adjustment. Most countries sit somewhere in between, because some spending categories are rigid (wages, pensions, debt service) while others can adjust (capital investment, discretionary programmes).

How to set it. Start from the composition of the budget:

- Countries with large public sector wage bills, universal social programmes, or rigid entitlement commitments: lean toward 0.7 to 1.0.
- Countries with more fiscal flexibility, smaller governments, or well-established expenditure review processes: 0.3 to 0.6.
- The default of 1.0 is deliberately conservative, showing the worst case where all climate damage becomes additional borrowing.

For detailed methodology, see the IMF User Guide, pp. 20 and 35-36 on the expenditure rigidity parameter and fiscal effects of climate change.

What the source shows. Rigidity is a claim about how spending has behaved, and the WEO record carries evidence on it.

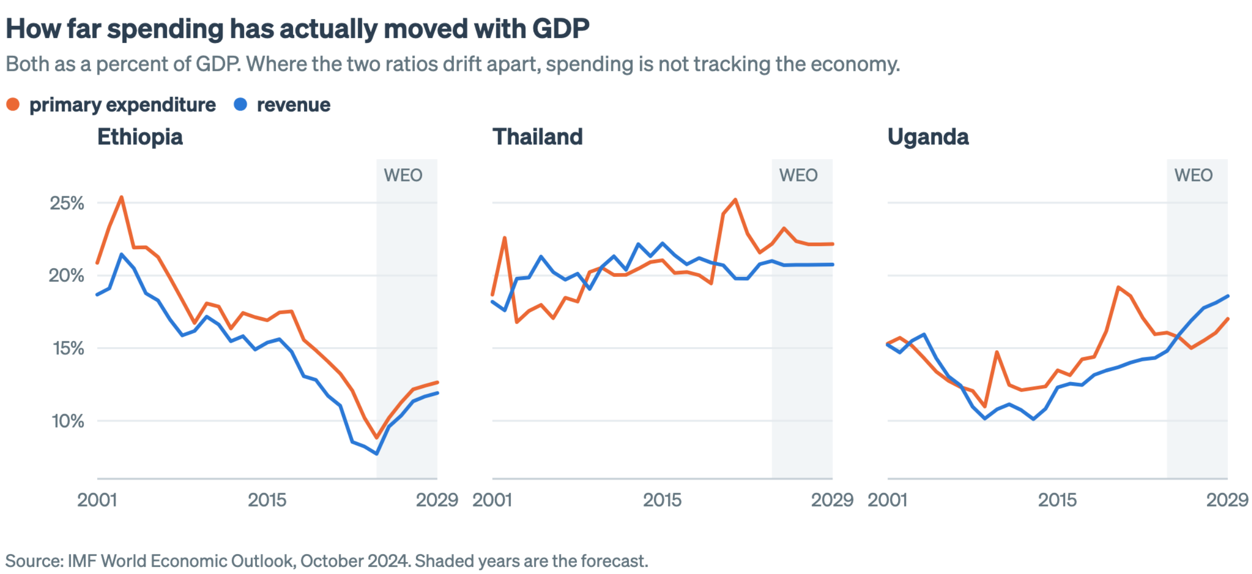


Figure 4.6: Rigidity is a claim about how these two lines move together. Where expenditure holds its share of GDP while revenue falls, the budget has behaved as though rigidity is high.

Read the two lines against each other rather than separately. Revenue holds a roughly constant share of GDP by construction in the projection, but in the historical record it moves, and the question is what expenditure did when it moved. A budget whose expenditure ratio climbs while revenue falls has behaved as though rigidity is high. That is evidence you can put in the packet, and it is stronger than a view about political will.

WIDGET-TODO: expenditure rigidity

Embed the expenditure rigidity intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). Iframe under Quarto for now, native React once the site moves to Astro. The integration pass happens after this restructure and run 3 both land, and this callout is the anchor point. The prose above must carry the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

Predict, observe, explain

Predict. On the Analysis tab you are about to compare the climate fan at rigidity 1.0 against rigidity 0.0. Write down which one produces the wider fan, and by roughly what factor.
Observe. Set rigidity to 1.0 and read the 2099 gap between the baseline and Hot Unadapted. Set it to 0.0 and read the same gap.
Explain. At 1.0 the fan spreads wide, because the whole climate revenue loss accumulates as debt, so the distance between Paris-aligned and Hot Unadapted is large. At 0.0 the scenarios compress toward the baseline, because the government absorbs the shock through spending. The distance between those two fans is your country's fiscal exposure to expenditure rigidity, and it is usually the largest single lever in the tool. If you predicted that rigidity matters less than the choice of climate scenario, check it, because for most countries it does not.

SCREENSHOT-TODO

Side-by-side Analysis tab for Uganda, the worked case, at rigidity 1.0 and rigidity 0.0, with the same axis limits on both, so the compression is visible rather than described. Same-axis is the point, because two auto-scaled charts will hide the effect.

Defend your choice: rigidity

This is the judgment call you will actually be challenged on. A colleague says 1.0 is alarmist: "no government holds spending flat in local currency for seventy years while the economy shrinks." Another says anything below 1.0 assumes a fiscal discipline your country has not demonstrated. What do you do?

> **DRAFT FOR TEAL: defending a rigidity choice**
> Both colleagues are making empirical claims, and the parameter is where you settle them with evidence rather than adjectives.
> **The evidence to reach for** is the composition of the budget, not a view about political will. What share of primary expenditure is wages, pensions, statutory transfers and other commitments that require legislation to change? That share is close to a floor on rigidity. In most low-income countries it is high, and the flexible remainder is dominated by domestically financed development spending, which is exactly the spending a government protects last.
> **The defensible position** is to run 1.0 as the headline and a lower value as the sensitivity, and to say why in one line: the default is conservative by design, and a fiscal risk statement is the document where conservatism belongs. Then give the second colleague their number, which is what the sensitivity run is for: at rigidity

0.6, the 2099 gap narrows from X to Y percentage points.
What not to do is pick a middle value because it feels balanced. An unexplained 0.5 is weaker than 1.0 with a stated rationale, because 1.0 at least has the tool's own conservative-default argument attached. A middle number with no reasoning behind it is the assumption that gets quietly reused for five years by people who never knew it was a guess.
The sentence for the packet: "Rigidity set at 1.0, the tool default, because roughly N percent of primary expenditure is wages, pensions and statutory commitments. The sensitivity run at 0.6 changes the 2099 climate-baseline debt gap from X to Y percentage points."

Document it

Record: **rigidity used**, **the budget-composition evidence behind it**, and **the sensitivity run and its effect on the headline gap**. This is the entry the capstone rubric weights most heavily.

4.7 Wrapper: what you can now do

- You can set all five controls for a country and give a reason for each that cites evidence rather than preference.
- You can predict each parameter's direction of effect before moving it, which is what tells you when the tool is behaving oddly.
- You have five **Document it** entries. Together they are the assumption-rationale section of your export packet.

Common errors on this material: inverting the rigidity scale (see Chapter 1, question 2), moving the debt target while the fiscal rule is off and wondering why nothing happened, and choosing a middle value to avoid an argument.

On your desk this week: find one number in a projection your department published and ask who chose it and why. If the answer takes more than a minute to find, you have made the case for the export packet.

Chapter 5

A worked example, end to end

In this module
You will run one country the whole way through, from parameter choices to publishable prose. The worked case is Uganda, because it is the tool's golden-master verification country and every figure in it can be checked against two published documents. The scaffolding then fades onto other countries: a completion problem on Ethiopia where you finish the interpretation, and an independent problem on a third country that you do alone. The target is not a chart but a write-up, in the format of the IMF's C-PIMA presentation of Q-CRAFT results and Section III of Uganda's Fiscal Risk Statement.

Fast path

There is no fast path through this module, because it is the module the capstone is built from. If you are short of time, do Section 5.6 and Section 5.8 and skip the middle.

5.1 Warm-up

From Chapter 4 and Chapter 2:

1. Which two parameters change the climate result but not the baseline?
2. In which year do climate scenarios begin to diverge from the baseline?

(Answers: expenditure rigidity, and the climate scenario itself. Everything else moves both. 2030.)

5.2 Two documents already show what the output looks like

You are not inventing a format. Two published documents show what Q-CRAFT output looks like when it reaches a policymaker. Both are about Uganda, because Uganda is where the tool has been through a full cycle from workshop to published statement. The genre is not Ugandan, and your ministry's version of it will look much the same.

The IMF's C-PIMA write-up. *Uganda: PFM Climate Assessment: Public Investment and Fiscal Risks Management* (IMF, 2024) reports the September 2023 workshop with staff of Uganda's

Ministry of Finance, Planning and Economic Development (MoFPED, the finance ministry) in a single paragraph. Climate change looks milder in Uganda than in other sub-Saharan African countries, it says, and it still matters for growth and fiscal sustainability, because by end of century GDP loss could pass 4 percentage points with debt at 66 percent of GDP against 47.5 percent in the baseline. One paragraph. Three numbers. A comparative judgment.

Uganda's Fiscal Risk Statement FY 2024/25. Section III, pages 13 to 17, is a fuller treatment: scenario definitions, a baseline fiscal path table, GDP deviation charts, fiscal balance and debt charts, and a closing policy sentence. The source line reads "QCRAFT (2023)."

Your capstone is the shorter of the two. This module builds it.

5.3 By the end of this module you can

- **Execute** a complete Q-CRAFT run for a country, from parameter selection to CSV export
- **Apply** the Baseline Sanity Check before interpreting any climate result
- **Interpret** the three chart families, saying what each one licenses you to claim
- **Draft** a two-paragraph fiscal risk write-up in the register of a published Fiscal Risk Statement
- **Assemble** the export packet: inputs, rationale, output

5.4 Where you are in the course

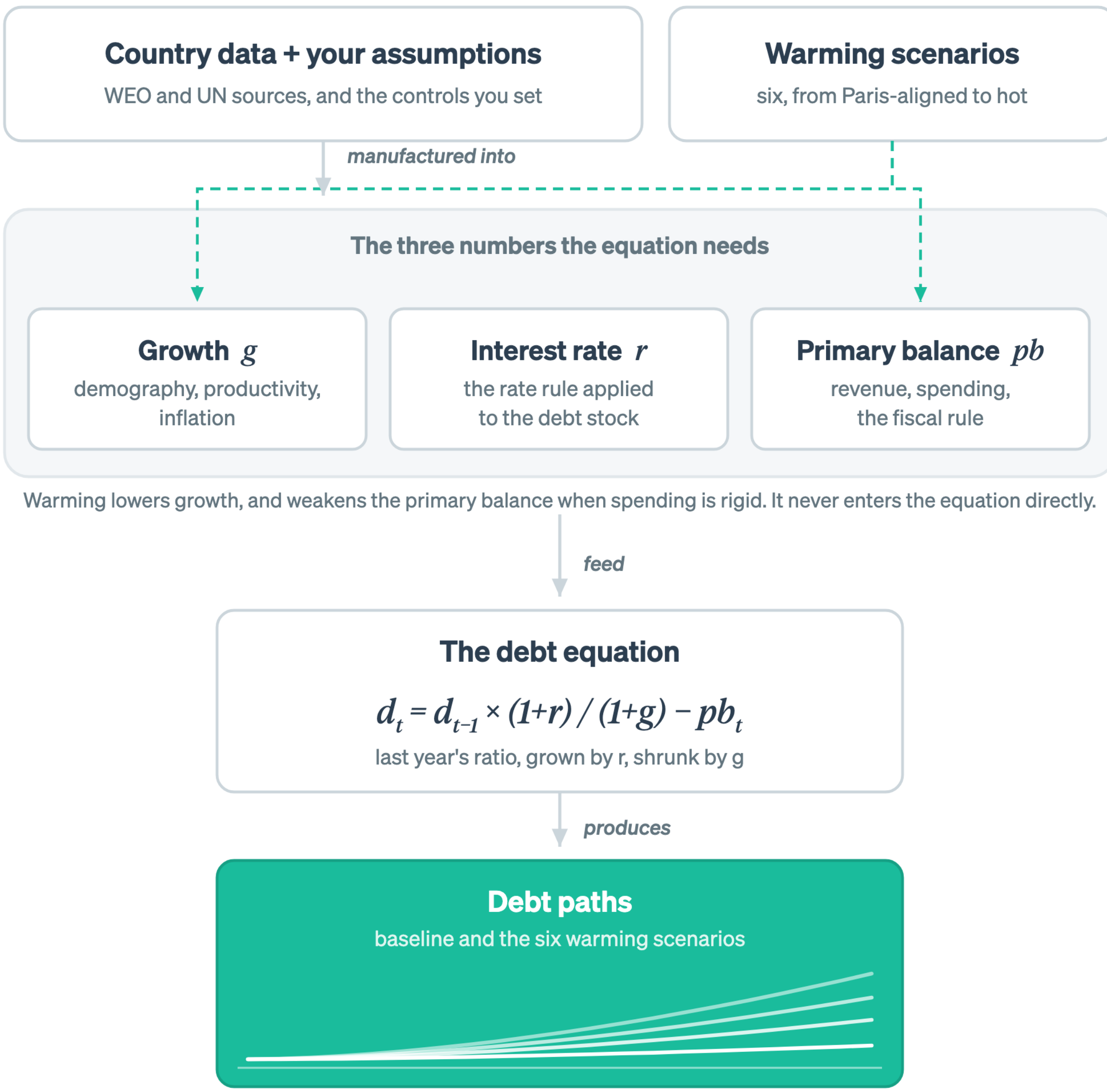


Figure 5.1: This module is the output end: the debt paths, and the write-up that comes out of them.

The last node is lit: the debt paths, and the write-up you build from them.

5.5 The target format

Section III of Uganda's Fiscal Risk Statement FY 2024/25 is organised in three moves. Copy that structure, because a ministry has already published it and it survived contact with its readers.

Move	What it contains	Where your material comes from
1. Set up the analysis	What the tool does, the scenarios used, and the baseline fiscal path	Chapter 4 parameters, Baseline tab
2. Macroeconomic effects	GDP deviation from baseline by scenario, and the productivity channel behind it	Climate tab
3. Fiscal effects	Primary and overall balance by scenario, debt-to-GDP by scenario, read against the fiscal anchor	Analysis tab

The published baseline table is worth copying too. Uganda's Table 5 reports four years, not seventy:

Summary fiscal path	2023	2050	2075	2099
Primary expenditure (% of GDP)	19.9	19.6	18.8	19.4
Interest expenditure (% of GDP)	3.3	1.3	1.8	2.4
Interest rate (%)	7.7	3.9	5.5	5.5
Primary balance (% of GDP)	-6.3	-1.7	-0.8	-1.4
Overall balance (% of GDP)	-9.6	-3.0	-2.7	-3.8
Debt-to-GDP ratio (%)	47.1	36.2	35.8	47.5

Source: Uganda Fiscal Risk Statement FY 2024/25, Table 5, sourced to QCRAFT (2023).

Four columns is a deliberate editorial choice, because a seventy-row table is not a policy document. Note what that baseline says before any climate damage is applied. Debt falls to the mid-30s by mid-century and comes back to 47.5 by 2099. Read the interest rate row against the growth path and you can see why.

5.6 The worked case: Uganda, start to finish

Uganda is the worked case because its numbers are published and checkable, not because it is the country you work on. The seven steps are the method, and the method is what transfers.

Seven steps follow, and you should follow along in the app. Every step names what you are doing and what it licenses you to say.

5.6.1 Step 1: set the parameters, and write down why

Open Q-CRAFT Explorer and select Uganda. Use these settings for the worked case:

Parameter	Setting	Rationale for the packet
Country	Uganda	Analysis subject
Demography variant	Medium	Reference variant, with Low tested as a sensitivity

Parameter	Setting	Rationale for the packet
Debt target	50% of GDP	The ceiling Uganda's own Fiscal Risk Statement reads results against
Fiscal rule	Off for the headline run	Discloses the risk without assuming the policy response, with the rule-on run reported alongside
Expenditure rigidity	1.0	Tool default, conservative, with budget composition evidence per Chapter 4 and 0.6 tested as a sensitivity

That table is the first page of your export packet. Write it before you look at any output, so the settings are choices rather than post-hoc justifications.

5.6.2 Step 2: read the Baseline tab

The Baseline tab shows the no-climate-change scenario: the country's fiscal trajectory under current trends and the assumptions you have set. Three charts share the tab.

Debt-to-GDP trajectory (top) is the headline chart. Follow the trajectory from the historical period, shaded through 2029, into the projection period. Is debt rising, stable, or falling? If the fiscal rule is on, debt should converge toward your target. If it is off, the trajectory reveals the underlying fiscal dynamics: favourable when growth exceeds interest rates, unfavourable when interest rates exceed growth.

Revenue and Expenditure (% of GDP) (bottom left) shows two ratios that can drift apart. Revenue-to-GDP stays constant by assumption, because it grows with nominal GDP. Primary expenditure-to-GDP may diverge, because expenditure grows with total population and productivity while revenue grows with working-age population. In countries where the working-age population grows faster than total population, which is the demographic dividend, expenditure-to-GDP falls and fiscal space opens. In ageing countries it rises, which is a structural fiscal pressure independent of any policy choice.

Fiscal Balances (bottom right) carries two lines: the primary balance, meaning revenue minus non-interest expenditure, and the overall balance, which includes interest payments. The zero line is the reference: a primary surplus means the government is generating enough non-interest revenue to cover non-interest spending. The gap between the two lines is the interest burden on existing debt.

> ⚠ SCREENSHOT-TODO
>
> Baseline tab for Uganda, all three charts, with callouts on: (a) the end of the shaded WEO period at 2029, (b) the 2099 debt ratio, (c) the primary-to-overall balance gap. This is the image the whole module refers back to.

5.6.3 Step 3: run the sanity check before you interpret anything

Baseline Sanity Check

Before interpreting climate results, verify the baseline makes sense:

- ☐ Does the initial debt level match your country's actual current debt-to-GDP ratio?
- ☐ Is the revenue-to-GDP ratio plausible given WEO forecasts?
- ☐ Does the expenditure path reflect realistic spending trends?
- ☐ If the fiscal rule is on, does debt converge toward your target?
- ☐ Do the primary and overall balance paths make economic sense given the country's fiscal history?

! DRAFT FOR TEAL: the sanity check applied to Uganda

Run the five boxes against what is on your screen and against the published figures.

Initial debt level. The published baseline puts 2023 at 47.1 percent of GDP, and Uganda's Fiscal Risk Statement reports 46.9 percent at June 2023 on the ministry's own basis. Those agree. If your Explorer run shows something materially different, the likely cause is the WEO vintage, because the Explorer ships October 2024 and the published table came from the 2023 run. Note the difference rather than fixing it by overriding data.

Revenue-to-GDP. Revenue is held at a constant share of GDP by construction, so the check is not whether it moves. It is whether the level it is held at is the right one. For a country in the middle of a revenue mobilisation strategy, a flat revenue ratio is a conservative assumption and worth one sentence in the write-up.

Expenditure path. The published table shows primary expenditure drifting from 19.9 to 19.4 percent of GDP over seventy-six years, dipping to 18.8 around 2075. That is the demographic dividend showing up: total population growth drives spending, working-age population growth drives GDP, and for a while the second outruns the first. If your expenditure path rises instead of drifting down, check the demography variant.

Fiscal rule convergence. With the rule off, skip this box, and say in your packet that you skipped it and why.

Balance paths. The published baseline has the primary deficit narrowing from 6.3 percent of GDP in 2023 to 1.4 percent by 2099, with the overall deficit at 3.8. That is a large near-term consolidation built into the WEO forecast period rather than something the model invented. Whether it is credible is a question about the WEO, and it belongs in your write-up as a caveat rather than a correction.

The rule this step encodes: a climate result computed on a baseline you have not checked is a number with no owner. Every one of the five boxes is a question someone will ask you.

5.6.4 Step 4: read the Climate tab

The Climate tab shows how climate change affects real GDP under six warming scenarios, using empirical estimates from the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on Kahn et al. (2021).

What to look for. Two charts show absolute GDP levels and a GDP index set to 100 in 2029. The six scenarios run from Paris-aligned, below 2°C warming by 2100, to Hot Unadapted, high warming with slow adaptation. Impacts begin in 2030, and before that year all scenarios match the baseline. The gap between scenarios is the range of GDP outcomes across different climate futures.

Countries closer to the equator generally show larger GDP losses. The empirical estimates reflect historical relationships between temperature deviations and economic output, and countries already in warm climates suffer more from additional warming.

A country showing a large gap between Paris and Hot Unadapted faces high climate-fiscal vulnerability. A small gap suggests lower direct climate exposure in the model, though it does not account for the indirect effects that Q-CRAFT's conservative methodology leaves out: natural disasters, sea-level rise, and trade disruptions (User Guide, p. 5).

> **!** DRAFT FOR TEAL: what the Uganda GDP chart licenses you to say
>
> The published result is a level GDP loss of around 4 percent by end of century under the Hot scenario, and the C-PIMA summary describes Uganda's impact as milder than other sub-Saharan African countries.
> Four percent needs a perspective sentence, because on its own it reads as small. Two are available and they point in opposite directions, so pick deliberately and say which you picked.
> The deflating one: 4 percent of GDP spread over seventy-five years is a few hundredths of a percentage point off annual growth, well inside the noise of any single year's outturn.
> The alarming one: it is a permanent level loss that compounds into the debt ratio every year thereafter, and 4 percent of GDP is a recurring annual amount comparable to a large sector budget.
>
> > **TODO: verify the perspective figure**
> > The second framing needs a checked comparator before it ships: 4 percent of Uganda's GDP expressed against a named line of the approved budget (health, or the road sector), sourced to the Approved Estimates for the relevant year. Do not ship an approximate comparison in a document aimed at the ministry that publishes the actual number.
>
> Both framings are true. The second is the one a fiscal risk statement is for, and the first is the one you should expect back from a sceptical reader. Write the paragraph so it survives both.
> **What the chart does not license:** any statement about what Uganda's GDP will be. This is a deviation between two runs of the same model, not a growth forecast.

> ⚠ SCREENSHOT-TODO
>
> Climate tab for Uganda, GDP index chart, with the six scenario lines labelled on the data rather than in a legend, and the 2030 divergence point marked.

5.6.5 Step 5: read the Analysis tab

The Analysis tab is the comparison view, overlaying baseline and all climate scenario debt trajectories on a single chart.

The climate-fiscal risk premium. The spread between the baseline debt trajectory and the climate scenario trajectories is the country's climate-fiscal risk premium: the additional debt burden attributable to climate change. Three factors set that spread.

1. **Climate exposure:** how much GDP the country loses under each scenario, from the Climate

tab.

2. **Expenditure rigidity:** how much of the GDP loss passes through to the primary balance and debt. Higher rigidity means a wider spread.
3. **Starting fiscal position:** countries with unfavourable baseline debt dynamics, where interest exceeds growth, see climate shocks amplified through the debt accumulation equation.

Reading the chart. A baseline showing stable or declining debt alongside a Hot Unadapted scenario that rises rapidly means climate change could destabilize an otherwise sustainable fiscal position. That is the core finding Q-CRAFT is designed to surface, and it answers one question: even if our fiscal policy is sound today, could climate change make it unsustainable?

> **!** DRAFT FOR TEAL: reading the Uganda fan chart
>
> Three readings, in the order a reviewer will want them.
> **The gap.** Baseline at 47.5 percent of GDP in 2099, Hot at 66. Eighteen and a half percentage points is the headline, and it is the number that goes in the first sentence. Uganda's own Fiscal Risk Statement says the same thing from the other end: the primary deficit in the Hot scenario ends 0.7 percentage points worse than baseline, "thereby raising public debt by over 18 percent of GDP." A 0.7 point annual deficit gap producing an 18 point debt gap is the compounding from Chapter 3 doing its work, and it is worth saying out loud, because it is the part non-specialists miss.
> **The threshold.** The gap only becomes a policy fact when it crosses something. Uganda's Statement reports that in the High, Hot and Vulnerable scenarios debt passes the 50 percent of GDP fiscal rule ceiling. That sentence does more work than the 18 points, because it converts a projection into a breach of an existing commitment.
> **The shape.** Look at when the lines separate, as well as where they end. Scenarios are identical until 2030, then fan slowly. A path that crosses the ceiling in 2085 and one that crosses in 2060 imply very different urgency, and the fan chart shows the difference at a glance where a 2099 table does not.
> **What to do when the baseline itself is rising.** If your run has debt rising in the baseline, the climate gap is no longer the interesting number, because both paths are unsustainable. Say so, and report the year each path crosses the anchor instead.

5.6.6 Step 6: export the packet

The Data tab holds an interactive data grid and two CSV downloads.

- **Download Baseline CSV:** the baseline fiscal projection, with no climate scenarios. Useful for folding Q-CRAFT projections into your own fiscal framework or DSA.
- **Download All Scenarios CSV:** baseline plus all six climate scenarios, stacked with a scenario identifier column. Use this for custom analysis, cross-country comparison, or integration with other tools.

All values are in billions of local currency units, except ratios, which are percentages of GDP. The data covers the full projection period from 2009 through 2099.

Download both. Add them to the parameter table from Step 1 and the Document it entries from Chapter 4. That is the export packet: inputs, rationale, output, in one place, reproducible by someone who has never met you.

5.6.7 Step 7: write the two paragraphs

! DRAFT FOR TEAL: model two-paragraph write-up

Built from the published Uganda figures so every number in it is checkable. Learners replace them with their own run.

> Climate change presents a measurable long-term risk to Uganda's fiscal position. Using the IMF's Quantitative Climate Risk Assessment Fiscal Tool, the baseline projection, which assumes no climate damage, has public debt at 47.5 percent of GDP in 2099 after falling to around 36 percent by mid-century as the demographic dividend widens the gap between growth and the effective interest rate. Under the Hot scenario, in which emissions follow SSP3-7.0 and temperatures track the 90th percentile of model projections, output is around 4 percent below baseline by the end of the century and public debt reaches 66 percent of GDP. The mechanism is indirect: higher temperatures slow labour productivity growth, slower growth reduces revenue, and with primary expenditure held at its baseline level in shilling terms the shortfall accumulates as debt. An annual primary deficit only 0.7 percentage points wider than baseline compounds into a debt ratio more than 18 percentage points higher over seventy years.
>
> The policy significance lies in the threshold rather than the magnitude. In the High, Hot and Vulnerable scenarios, public debt passes the 50 percent of GDP fiscal rule ceiling and takes on a rising trajectory, which means climate damage alone is sufficient to move Uganda from compliance to breach without any other adverse shock. These estimates are conservative. They capture the slow productivity effects of temperature change and exclude natural disasters, sea-level rise, non-market damages and the public expenditure required to adapt, so the true fiscal exposure is larger than reported here. The results are also sensitive to the expenditure rigidity assumption: at full rigidity, the tool's default, the entire revenue loss becomes borrowing, while at 0.6 the end-century gap narrows materially. Meeting Uganda's Paris commitments and building expenditure flexibility both reduce this exposure, and the two are complements rather than alternatives.

Six things make this a Fiscal Risk Statement paragraph rather than a report of a model run:

1. It names the tool and the scenario in one clause each, then moves on.
2. Every number is a comparison between two runs, never a forecast.
3. It gives the mechanism in one sentence, so a reader who does not know the debt equation can still follow the causal chain.
4. It states the threshold, because a breach of an existing commitment is what makes a projection actionable.
5. It states the conservatism caveat before anyone else can, which is what makes the rest credible.
6. It ends on something the ministry can do.

One editorial query for Teal: the closing sentence of the second paragraph does policy advocacy inside a risk document. Whether that register is right is a call I should not make on your behalf.

Now do your own country

The seven steps are a method, and the method is the part that transfers. Uganda supplied numbers you could check it against. Nothing in the sequence depended on the country: set the parameters and write down why, check the baseline before you interpret anything, read the three tabs in order, export the packet, write two paragraphs.
The rest of this module runs that method twice more, on countries that behave differently from Uganda and from each other, with less help each time. The capstone in Chapter 7 runs it on yours.

5.7 Completion problem: Ethiopia under the High scenario

New country, same method, most of the setup fixed for you. Ethiopia earns its place here because its demography runs the other way from most countries in a fiscal risk conversation: the working-age population is still climbing, which pushes the baseline in one direction while climate damage pushes the other.

Done for you:

- The method: the same seven steps, in the same order. You are not inventing a procedure.
- The scenario: **High** is SSP3-7.0 at the average of model temperature projections, rather than the 90th percentile.
- The parameter template from Step 1, with one open cell. The debt target is Ethiopia's own fiscal anchor if you can establish one, and 50 percent of GDP if you cannot. Record which you used and why.

Your turn, six tasks:

1. Run the Baseline Sanity Check on Ethiopia. This is a different baseline from the worked case, so the box is now yours and it comes before any climate reading.
2. Read the 2099 debt ratio under High from the Analysis tab. Record it. ______
3. Compute the gap to baseline. ______ percentage points.
4. Read the year the High path crosses your target, if it does. ______
5. Write one sentence explaining why High sits below Hot when both use the same emissions pathway.
6. Rewrite the first sentence of the model paragraph above using your Ethiopia numbers.

DRAFT FOR TEAL: checking yourself on task 5

The distinction is the temperature distribution rather than the emissions. High takes the average temperature outcome across climate models running SSP3-7.0, and Hot takes the 90th percentile of that same set. Same policy world, different draw from the uncertainty. Which one belongs in a fiscal risk statement is a real question: risk documents conventionally report adverse tail outcomes, which argues for Hot, while a central-case fiscal framework argues for High. Report both and say why you led with the one you led with.

5.8 Independent problem: adaptation speed, on a third country

No scaffolding this time, a checklist only, and a country whose fiscal position looks like neither of the first two.

The task. Compare **Hot + Adapted** against **Hot + Unadapted** for **Zambia**, or for another country in the dropdown whose interest-growth differential runs against it. Temperatures are the same in both scenarios, and the difference is how quickly the economy adjusts.

Deliver five things:

- ☐ The 2099 debt ratio under each, and the gap between them
- ☐ One sentence on what that gap represents in policy terms
- ☐ One sentence on what it does **not** represent, which is the trap in this comparison
- ☐ The same comparison at expenditure rigidity 0.6, with a note on whether the adaptation gap or the rigidity gap is larger for your country
- ☐ A two-paragraph write-up in the format of Step 7

> **i** DRAFT FOR TEAL: the trap, if you want to check before you write
>
> Faster adaptation in Q-CRAFT is a faster economic adjustment to a given temperature path. It is not adaptation spending. The model does not cost the investment that would produce that adjustment, and the User Guide says so directly (p. 6). So the Adapted-to-Unadapted gap is the benefit side of an adaptation programme with the cost side missing. Reporting it as the value of adaptation without that sentence attached is the likeliest way to misuse this tool in a budget discussion.

5.9 Wrapper: what you can now do

- You have run three countries, one with full support and one alone, and produced an export packet with rationale attached.
- You can apply the Baseline Sanity Check and say what each box protects you from.
- You can write two paragraphs that a Fiscal Risk Statement editor would recognise.

Common errors on this material: interpreting climate results before checking the baseline, quoting a 2099 level as a forecast, and reporting the adaptation gap as the value of adaptation.

On your desk this week: take the two paragraphs you drafted to whoever owns your ministry's fiscal risk chapter and ask what they would cut. The answer tells you more about the format than this module can.

Chapter 6

What the tool can and cannot tell you

> **In this module**
> You will start with what the tool does well and why the IMF built it that way, because a caveat is only useful to someone who knows what it qualifies. Then you will learn what Q-CRAFT leaves out and which direction the omissions run, so you can state the limit before a reviewer states it for you. You will learn to read a fan chart whose baseline is clipped at zero when the climate scenarios are not. Last, you will practise telling a Q-CRAFT question from a debt sustainability question, in both directions.

> **Fast path**
>
> No fast path. This is the shortest module and it is the one that keeps you out of trouble. If you only read one section, read Section 6.7.

6.1 Warm-up

From Chapter 5:

1. What does the gap between Hot Adapted and Hot Unadapted represent, and what does it leave out?
2. Which check comes before any climate interpretation?

(Answers: the benefit of faster adjustment, with the cost of achieving it not modelled. The Baseline Sanity Check.)

6.2 The question that comes after the presentation

You have presented the fan chart. The senior official who will have to defend it looks at the Hot Unadapted line and asks: "So this is what climate change will cost us?"

Your whole credibility rests on the next thirty seconds. This module is that answer, in its several parts.

6.3 By the end of this module you can

- **Name** four things Q-CRAFT does well, and **say** which design choice buys each one
- **List** the specific damage channels Q-CRAFT excludes, and **state** the direction of the resulting bias
- **Explain** why baseline and climate scenarios treat negative debt differently, and **adjust** your reading of a chart accordingly
- **Compare** two countries with similar debt paths and **distinguish** the different drivers behind them
- **Decide** whether a given question calls for Q-CRAFT, the LIC-DSF, both, or neither, and **justify** the choice

6.4 Where you are in the course

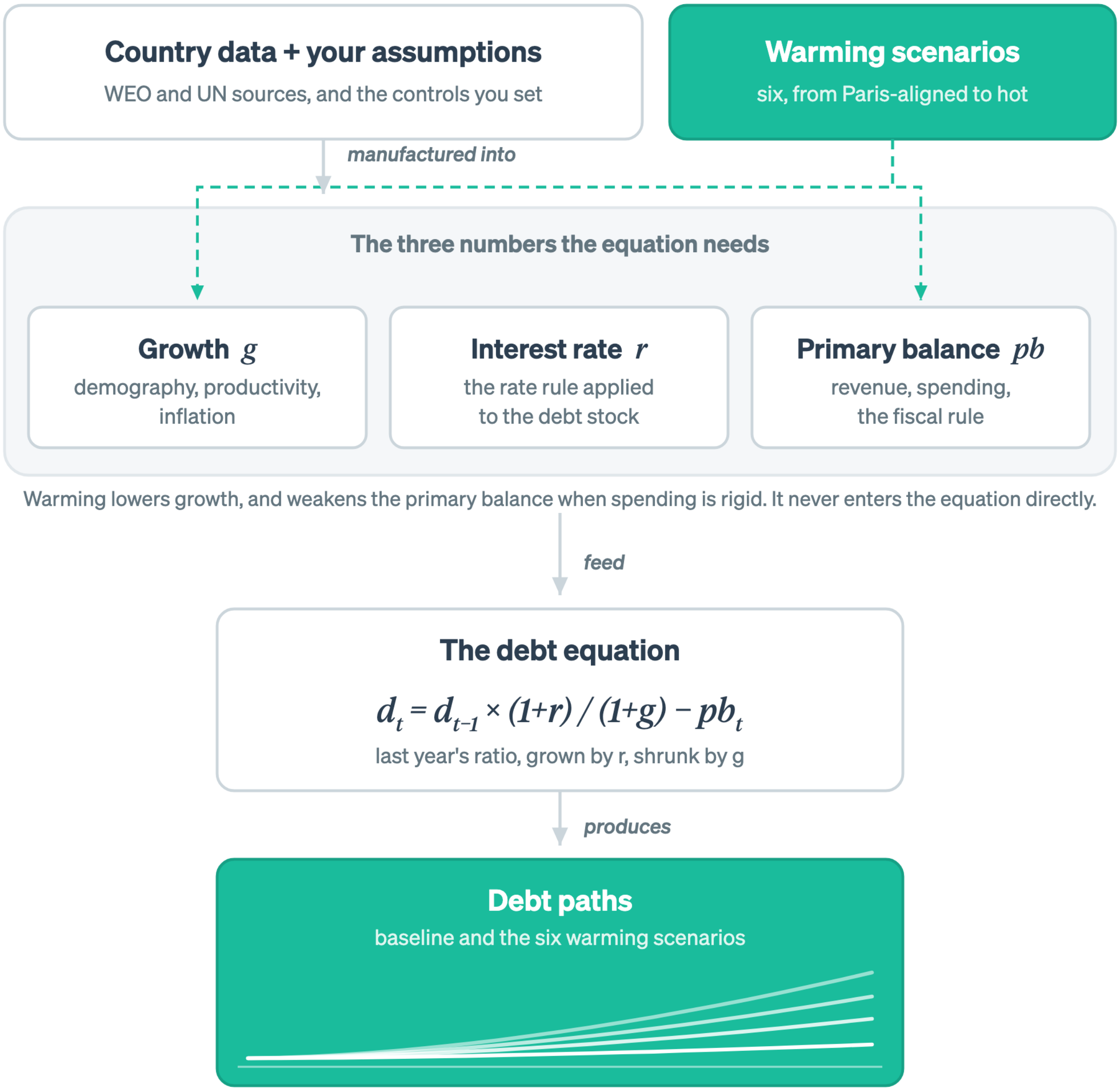


Figure 6.1: This module is about what the warming scenarios contain, and what the debt paths therefore license you to write.

Two nodes are lit, and they are the two ends of the climate channel: what the warming scenarios contain, and what the debt paths let you write.

6.5 What the tool is good at, and why it was built that way

Q-CRAFT does four things well, and each one is a design choice with a cost attached. The rest of this module is those costs, so the strengths come first.

It isolates one channel and models it from data. Temperature affects growth, growth affects revenue, and revenue affects debt. Q-CRAFT takes the temperature-to-growth step from cross-country empirical estimates using the Kahn et al. (2021) methodology, updated in the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), then carries it through a standard debt equation. One channel, estimated from data, pushed through arithmetic a reviewer can check line by line. That is a narrower claim than an integrated assessment model makes, and it is a claim you can defend in a room.

It runs on data every country already has. The User Guide is explicit that the tool can be deployed "without any additional data, even in low-capacity settings", building its scenarios from IMF, UN and World Bank sources (p. 6). No new collection, no survey, no waiting for a mission. A team can have a first pass by the end of the week. The same page is equally explicit that country expertise is still needed to make the baseline match the country's own fiscal framework, which is what Chapter 4 is for.

It is comparable across countries. The same production function, the same debt equation and the same six scenarios run for every economy in the tool. Two countries' results can therefore be set beside each other, which is what makes the diagnosis exercise later in this module work at all.

It produces the shape a fiscal risk chapter needs. The output is a comparison between scenarios rather than a forecast, and a risk chapter is a comparison between scenarios. The User Guide positions Q-CRAFT as "a starting point for climate change fiscal risk analysis" and as a foundation for deeper work with general equilibrium models where capacity allows (pp. 5-6).

Those four follow from one brief. The Fiscal Affairs Department needed something a country team could run in a week, on data that already exists, producing a defensible comparison rather than a false forecast. Every limitation in the rest of this module follows from the same brief, which is why they read as costs of a choice rather than as oversights.

6.6 What the numbers mean, and what they do not

Q-CRAFT Explorer produces stylized long-term projections. They are useful for understanding the direction and magnitude of climate-fiscal risks, comparing scenarios, and identifying the fiscal parameters that matter most for a given country.

The results are not forecasts. Projections to 2099 depend on assumptions about productivity, inflation, interest rates, and climate that involve deep uncertainty. The value sits in the comparison across scenarios and the sensitivity to assumptions, never in any single projected number.

Q-CRAFT is designed to complement existing fiscal analysis rather than replace it. The IMF User Guide positions it as "a starting point for climate change fiscal risk analysis" that supplements country macroeconomic models (User Guide, pp. 5-6). Use Q-CRAFT to identify which climate-fiscal channels matter most for your country, then investigate those channels with more detailed models and country-specific data.

6.7 Every exclusion is a cost left out, so the estimate is a floor

The scenarios capture one channel: the slow effect of temperature change on productivity, and through it on growth. Six things are excluded, the User Guide names all six, and every one of them runs in the same direction.

Excluded	Documented at	Why it matters for a fiscal risk statement
Climate-induced natural disasters	User Guide, p. 5	Often the channel a national risk statement leads with. Uganda's puts floods and epidemics at about 75 percent of recorded events, 1985 to 2021
Sea-level rise	User Guide, p. 5	Absent for a landlocked economy, decisive for Bangladesh or a small island state
Tipping points	User Guide, p. 5	The estimates are fitted to historical temperature variation and cannot extrapolate to discontinuities
Non-market damages	User Guide, p. 5	Morbidity, mortality, conflict, food insecurity: real costs, absent from GDP
Spillover effects	User Guide, p. 5	Trade partners, remittance sources and commodity markets are hit too
Adaptation spending	User Guide, p. 6	Even the Hot Adapted scenario models faster adjustment, not the public expenditure that buys it

The direction is what makes this list usable. Every exclusion is a cost left out, so the modelled fiscal impact is a lower bound on the modelled channels rather than a central estimate of total impact. The User Guide reaches the same conclusion in its own voice: the results "do not account for the potential impacts of climate change induced natural disasters, sea-level rise risks and other environmental risks, rendering the outcomes conservative" (p. 5). One sentence follows from it: *actual fiscal impacts are likely larger than these results suggest.*

Say it yourself, in the write-up, before a reviewer says it for you. A conservative number with the conservatism disclosed is stronger than a larger number without it.

Some practitioners judge the damages conservative beyond that list. Their argument is about the estimates rather than about what surrounds them. The empirical estimates are fitted to historical variation in temperature, so they carry information about the range the world has already lived through and none about the range it has not. Whether that understates the damage from warming well outside the historical range is an open empirical question. Expect to be asked it, and answer it the way this module keeps answering things: report the number, name the channel it covers, and name the channels it does not.

Partial equilibrium is a limit of a different kind. The User Guide describes the set-up as "essentially a partial-equilibrium" one (p. 5) and says plainly that Q-CRAFT "is not a forecasting model nor a general equilibrium model of the economy" (p. 6). Fiscal policy does not feed back into growth or interest rates, so cutting spending inside the model never slows the economy down. That limit has no obvious sign, unlike the six above: it flatters a consolidation scenario, and it also leaves out whatever growth an adaptation programme would buy. What it means in practice is that Q-CRAFT cannot cost a policy. It can show you what happens to the debt path under assumptions about policy that you supplied, and that is a different thing.

> **!** DRAFT FOR TEAL: what conservatism does not buy you
>
> Conservatism protects the credibility of a number. It does not make the number safe to quote out of context, and it creates one specific hazard worth naming.
> A conservative estimate that gets picked up as *the* estimate becomes a ceiling in someone else's argument. "The analysis shows climate costs us 18 points of debt" is a sentence that starts life as your careful lower bound and ends life as a budget office's reason to fund adaptation at 18 points' worth. The defence is structural rather than rhetorical: put the exclusion list in the same paragraph as the headline number, not in a footnote, and not in an annex.

6.8 The baseline floors debt at zero and the climate scenarios do not

> **i** The rule
>
> The baseline scenario applies a floor of zero to the debt-to-GDP ratio: if the equation produces a negative value, debt is set to zero. Climate scenarios do not apply this floor, and debt is allowed to go negative, representing net asset positions. The asymmetry is intentional, because it avoids masking the full range of climate-scenario outcomes.

> **!** DRAFT FOR TEAL: how the asymmetry changes what you are looking at
>
> The asymmetry matters for a narrow set of countries, and for those countries it matters a lot.
> **When it bites.** Only when a projection drives debt to zero. That happens for countries with strongly favourable interest-growth differentials and sustained primary surpluses over a seventy-year horizon. It is not a problem in the worked case, whose published baseline lands at 47.5 percent of GDP.
> **What it does to the picture.** In such a country the baseline flattens along zero while the climate scenarios continue downward into negative territory. Read naively, the chart shows climate scenarios with *lower* debt than baseline, which is nonsense. What you are actually seeing is one line clipped and the others not.
> **The reading rule.** If any line in your fan chart touches zero, stop using the vertical gap as your headline number. Check the underlying series in the Data tab, then either report the comparison over a window before the floor binds or report the primary balance gap instead, which is not clipped.
> **Why the asymmetry is a design choice rather than a bug.** Flooring the climate scenarios too would compress exactly the range the tool exists to show. The choice preserves the spread and pushes the interpretive burden onto you. This module is where that burden gets paid.

6.9 Two countries, same chart, different problem

Transfer is the thing this course is actually for: recognising which situation you are in when the labels come off. Structured comparison is how it gets taught.

Comparison exercise: same path, different drivers

Find the pair. In the Explorer, work through countries in the dropdown until you find two whose baseline 2099 debt ratios are within about 5 percentage points of each other, but whose baseline-to-Hot-Unadapted gaps differ by 10 percentage points or more.
Then diagnose. For each country, decide which of the three drivers from Chapter 5 explains the difference:

1. Climate exposure, meaning how much GDP is lost, from the Climate tab
2. Expenditure rigidity, meaning how much of that loss becomes debt
3. Starting fiscal position, meaning whether the interest-growth differential amplifies or damps it

Write one sentence per country saying which driver dominates, and one sentence saying what policy conclusion follows for each. The conclusions should differ. If they do not, you have picked a pair that is too similar.
Then reverse it. Find two countries with similar climate exposure on the Climate tab whose debt outcomes diverge. The driver is now somewhere in the fiscal position, and finding it is the exercise.

TODO: seed the comparison with verified pairs

This exercise currently sends learners hunting. Two or three verified country pairs, run and checked against the current data vintage, would make it usable in a 20-minute workshop breakout. Needs a real run rather than an assumption about which pairs work.

DRAFT FOR TEAL: why this exercise and not a worked comparison

The instinct is to hand over the pair and the answer. Doing the search is the exercise, because the skill being built is diagnosis from surface features that mislead, and a country pair you were handed has already had the diagnosis done for you.
The general principle is the one to carry out of this module: a debt path is an outcome, not a diagnosis. Two countries arriving at 60 percent of GDP in 2099, one because it is highly exposed to warming and one because it borrows at rates above its growth, need different policies and different chapters in different documents. Q-CRAFT will show you the outcome. Deciding which country you are looking at is your job, and it is the part that cannot be automated.

6.10 Q-CRAFT and the LIC-DSF answer different questions

Q-CRAFT and the IMF-World Bank Low-Income Country Debt Sustainability Framework both produce debt projections for low-income countries. They are not substitutes, and using one where the other belongs is a common and expensive error.

	Q-CRAFT	LIC-DSF
Question it answers	How much does climate damage change the long-run fiscal path?	Is this country's debt sustainable, and at what risk rating?
Horizon	To 2099	Medium term, with longer-run tails
Output	Scenario comparison, no rating	A risk rating that carries operational consequences
Climate	Central: six scenarios	Not the organising principle
Debt detail	One aggregate stock	External and public splits, currency, creditor composition
Used for	Fiscal risk disclosure, C-PIMA annexes, long-term planning	Lending decisions, concessionality terms, programme design

Self-check: five questions, which tool?

For each, name Q-CRAFT, LIC-DSF, both, or neither, and say why in one line.

1. "The Fiscal Risk Statement needs a section on climate. What do we run?"
2. "Are we still at moderate risk of debt distress?"
3. "If we take this Eurobond instead of an IDA credit, what happens to our risk rating?"
4. "What is the fiscal cost of the National Adaptation Plan?"
5. "Does climate risk change our debt sustainability assessment?"

DRAFT FOR TEAL: self-check answers

The test in every case is which question the tool was built to answer, never which tool you happen to have open.

1. **Q-CRAFT.** This is its designed use, and it is the use Uganda already made of it in the FY2024/25 statement.
2. **LIC-DSF.** Q-CRAFT produces no risk rating and has no thresholds to breach. Answering this question from a Q-CRAFT chart is the error this section exists to prevent.
3. **LIC-DSF.** The question is about creditor composition and terms, which Q-CRAFT does not represent, because it carries one aggregate debt stock and one effective interest rate.
4. **Neither.** Q-CRAFT does not cost adaptation, by design (User Guide, p. 6), and the LIC-DSF does not either. This question needs a costed adaptation plan, and the honest answer is to say so rather than to produce a number from a tool that cannot see it.
5. **Both, in sequence, and this is the interesting one.** Q-CRAFT establishes whether the climate channel is large enough to matter for the fiscal path. If it is, the finding belongs alongside the DSF assessment as a qualitative risk consideration rather than as an input to it, because the two run on different horizons and different definitions of debt. Anyone who tells you they have mechanically fed Q-CRAFT output into a DSF has done something that needs explaining.

The reverse direction is the part usually skipped. A question that arrives framed for the DSF may be a Q-CRAFT question. "Our risk rating is fine, so climate is not a debt issue" is a DSF answer to a long-horizon question the DSF does not ask. Uganda's own result makes the point: a baseline that is comfortable on any medium-term view still crosses the fiscal rule ceiling by end of century once climate damage is applied.

6.11 Wrapper: what you can now do

- You can say what the tool does well and which design choice buys each strength.
- You can list what the scenarios exclude and state the direction of the bias without looking it up.
- You can spot a clipped baseline in a fan chart and say why the vertical gap has stopped meaning what it usually means.
- You can take a question and say which tool it belongs to, in both directions.

Common errors on this material: quoting a conservative estimate as a total, reading a floored baseline as a favourable climate result, and answering a sustainability-rating question from a Q-CRAFT chart.

The thirty-second answer, for reference: "No. It is what climate change does to the debt ratio through one channel, growth, under stated assumptions about how spending responds. It excludes disasters, sea-level rise and the cost of adapting, so it is a floor rather than a total. What it tells us is that the channel is large enough to breach the ceiling on its own."

On your desk this week: find a projection in circulation in your ministry and write down one question it cannot answer. Circulate that sentence with it.

Chapter 7

The capstone

In this module
You will produce the capstone: an export packet plus a two-paragraph fiscal risk draft, for a country you choose. It is marked on whether you can defend your assumptions, never on whether your numbers match anyone else's. The module also closes the loop on Chapter 1, so you can see what changed.

Fast path

Short on time? Read the brief and the rubric, then do the work. The retake at the end takes ten minutes and is the cheapest way to find out whether the economics landed or only the clicking did.

7.1 Warm-up

From across the course, without looking:

1. Which two of the three numbers does climate damage move?
2. What does expenditure rigidity 1.0 mean?
3. Name one thing the climate scenarios exclude.

(Answers: g and pb. Spending is sticky, so the revenue loss becomes borrowing. Natural disasters, sea-level rise, tipping points, non-market damages, spillovers, or adaptation spending.)

7.2 By the end of this module you can

- **Produce** a complete export packet for a country of your choosing
- **Defend** each parameter choice against a specific challenge
- **Assess** your own draft against the rubric before anyone else sees it
- **Compare** your current capability against your Chapter 1 self-assessment

7.3 Where you are in the course

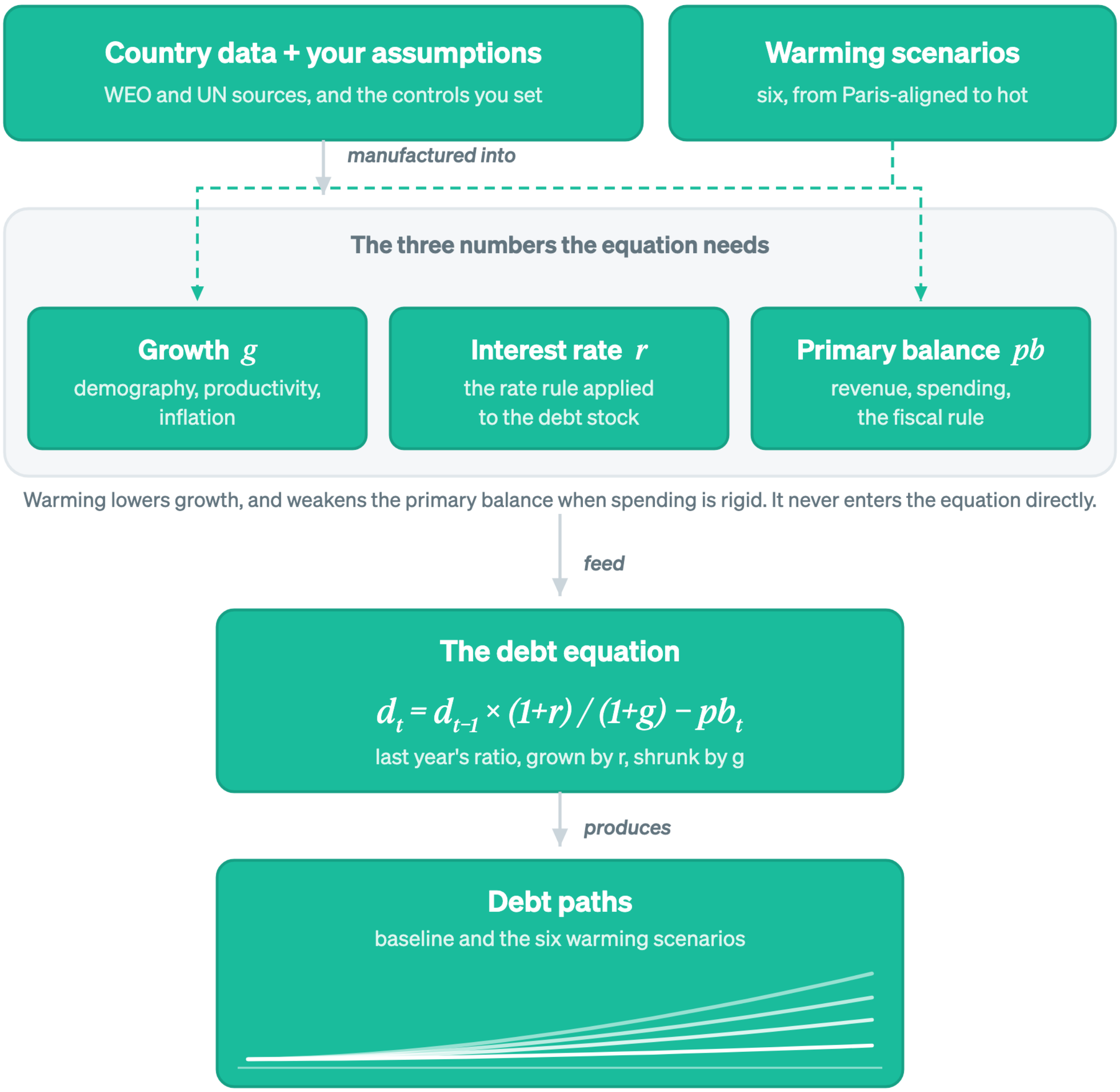


Figure 7.1: The capstone runs the whole chain, from the data you load to the paths you hand over.

Every node is lit, because the capstone runs the chain end to end.

7.4 The capstone brief

The scenario. You have been asked for a climate-fiscal risk section for your ministry's next fiscal-risk documentation, and you will defend it to the senior officials who sign it off. You have the Explorer, the WEO vintage it ships with, and whatever you know about the country. You have one week and half a page.

What to hand in, in three parts.

1. **The export packet.**
   - The parameter table: every setting, with a one-line rationale each (the five **Document it** entries from Chapter 4)
   - The data vintage, stated
   - Both CSV exports, baseline and all scenarios
   - At least two sensitivity runs, with what changed and by how much
2. **The two-paragraph draft**, in the format of Chapter 5, Step 7.
3. **Three challenge answers.** Written responses of two to three sentences each to:
   - "Is this a forecast?"
   - "Why did you choose that rigidity value?"
   - "So this is what climate change will cost us?"

Country choice. Pick the country you work on, because the capstone is a rehearsal for the document you will actually have to write. Pick Uganda instead if you want a run you can compare against the published Section III. Avoid Ethiopia and Zambia unless you want to extend Chapter 5, since you have already worked those.

What is not required. Matching anyone else's published figures. You are on a later WEO vintage and possibly different parameters, and reproducing someone else's numbers is not the skill being assessed.

7.5 The rubric

! DRAFT FOR TEAL: capstone rubric

Four criteria, and the weights are a proposal rather than a decision.

Criterion	Weight	Meets the standard	Falls short
Assumptions defended	40%	Every parameter has a rationale citing country evidence. Rigidity in particular has budget-composition reasoning behind it, and a sensitivity run.	Parameters listed without reasons, or reasons that restate the setting ("rigidity 1.0 because spending is rigid").
Baseline checked	20%	The five sanity-check boxes are answered against the actual run, including any that were skipped and why. Differences from the ministry's own figures are explained rather than reconciled away.	Climate results interpreted with no evidence the baseline was examined.

Interpretation	25%	Numbers presented as comparisons between runs. The mechanism stated in one sentence. A threshold identified. The conservatism caveat in the same paragraph as the headline number.	A 2099 level quoted as a prediction, a debt gap described as a cost, or the caveat relegated to a footnote.
Written for the reader	15%	Reads like your ministry's own risk chapter. A non-specialist can follow the causal chain. Two paragraphs, not five.	Reads like a description of a model run. Jargon undefined. Over length.

One error disqualifies on its own, and it is worth naming separately because it recurs: reporting the Adapted-to-Unadapted gap as the value of adaptation. It is the benefit side with the cost side missing (Chapter 5, independent problem).

Note for the live workshop. The rubric doubles as the peer-review instrument, run in three passes. First pass, each reviewer marks assumptions only. Second pass, interpretation only. Third pass, the writing. Splitting the passes stops reviewers spending all their attention on prose, which is what happens when they are asked to review everything at once.

7.6 Common errors to watch for

⚠ TODO: replace with pilot-observed errors

The list below is built from the misconceptions this course was designed against, not from watching people make them. The redesign plan treats a pilot with two or three target-profile testers as non-negotiable, and the error log from that pilot should replace this list. Until then, treat it as a hypothesis.

- Reading rigidity backwards, so the headline risk number comes out inverted
- Interpreting climate results before checking the baseline
- Quoting a 2099 level as a forecast
- Adjusting the debt target while the fiscal rule is off, then concluding the target does nothing
- Describing a debt-ratio gap as a cost in local currency
- Reporting the adaptation gap as the value of adaptation
- Answering a debt-sustainability-rating question from a Q-CRAFT chart

7.7 Where you started, where you are

Go back to your three answers from Chapter 1. Answer them again now, before reading on.

On long-term fiscal projection: A1 (not worked with a DSA) / A2 (can follow one) / A3 (could build one this week)

On climate-fiscal analysis: B1 (climate side only) / B2 (can describe the use) / B3 (have run or interpreted one)

On the tool itself: C1 (never opened it) / C2 (clicked around) / C3 (ran it and explained the output)

The A and B rows are the ones worth looking at. If you have completed the capstone, C3 is true by construction, because you have run the tool and written the explanation. Movement on A and B is the course working. No movement on A and B with C3 achieved means you can drive the tool while the economics underneath it did not land, and that is worth a conversation rather than a shrug.

Retake the three questions from Module 0

Same three items, unchanged. Answer from memory before opening anything.

1. Zero primary balance, $r = 9$ percent, $g = 6$ percent. Debt ratio next year: falls, flat, or rises?
2. Expenditure rigidity 1.0 means what about spending under a climate scenario?
3. Debt at 66 percent under Hot against 47.5 percent baseline. What is the defensible statement of what that means?

   DRAFT FOR TEAL: the answers, and what a wrong one means now

   A wrong answer here costs more than it did in Chapter 1, because now it is attached to a document you are about to circulate.

   1. **Rises**, to about 51.4 from 50. If this one is still wrong, Chapter 3 did not do its job and nothing in your capstone's interpretation section can be trusted, because every climate result in the tool is this arithmetic with a lower g.
   2. **Spending stays at its baseline level in local currency terms**, so the whole revenue loss becomes borrowing. Getting this wrong at the end of the course means your headline number may have the wrong sign on its sensitivity, which is the error most likely to reach print.
   3. **Under stated assumptions, climate damage of this magnitude adds roughly 19 percentage points of GDP to the debt ratio by 2099 relative to the same projection without it.** A comparison between two runs, with the assumptions attached. If you reached for the forecast framing or the cost framing, reread Chapter 6 before you circulate anything.

7.8 What to use on your desk this week

- The debt-stabilizing primary balance for your country: three numbers, one division, and it opens most credible fiscal risk write-ups.
- The **Document it** habit: every assumption gets a line saying why. It costs a minute per parameter and it is the difference between analysis and a spreadsheet nobody can defend a year later.
- The which-tool question. When a request arrives, ask what question it is asking before opening anything.

7.9 Spaced follow-up

Two things arrive after the workshop, and they are part of the course rather than an afterthought.

- **Day 3:** one retrieval question by email. No preparation, no marking.
- **Week 2:** an application check. What did you run, what did you find, what got used.

Spacing beats massing, and the IMF's own evaluation of capacity development found that training with hands-on follow-up sticks while the standalone workshop decays. The follow-up is the part that makes the rest count.

> ⚠ TODO: workshop materials
>
> The live sessions need artefacts this guide does not yet contain: breakout task cards, the anonymous poll items, and the peer-review sheet derived from the rubric above. Tracked separately from the book.

7.10 Wrapper: the whole course in six lines

- One equation decides the debt path, and the rest of the tool builds its three inputs. (Chapter 2)
- Two things move the ratio: the interest-growth gap, and the primary balance. (Chapter 3)
- Every parameter is a judgment call, and the record of the call is the deliverable. (Chapter 4)
- The output is a comparison between two runs, written up in your ministry's own format. (Chapter 5)
- The estimate is a floor, the baseline is floored at zero and the scenarios are not, and some questions belong to a different tool. (Chapter 6)
- What you hand over is an analysis you can defend. (this module)

Appendix A

Appendix: from Q-CRAFT to the LIC-DSF

> **This appendix is not part of the course**
>
> The modules teach you to run the tool and defend the analysis. This appendix is about the project behind the tool: why it was built, what it is a proof of concept for, and how to shape the next version. Nothing in the capstone depends on it. Read it if you want to influence what V2 does.

> **What you need to know:**
> - Q-CRAFT is a proof of concept for a larger goal: making the LIC-DSF more usable through human-centered design
> - The biggest barrier to effective fiscal projection tools is ergonomics, not economics
> - We are looking for co-design partners, not endorsements

A.1 The real barrier: ergonomics, not economics

The biggest barrier to effective use of fiscal projection tools is not the economics. It is the ergonomics. Smart people struggle not because the methodology is too complex, but because the tools do not guide them through the decisions they need to make.

This observation has emerged from years of conversations with MoF economists, central bank staff, and IFI economists. It has inspired a more organized human-centered design approach: structured design sprints and organized discussions with practitioners who use these tools in their daily work.

A.2 What we built and what we proved

Q-CRAFT Explorer reimplements the IMF's Q-CRAFT Excel workbook as an open-source Python web application. The calculation engine is a standalone package with automated parity tests against the original Excel tool.

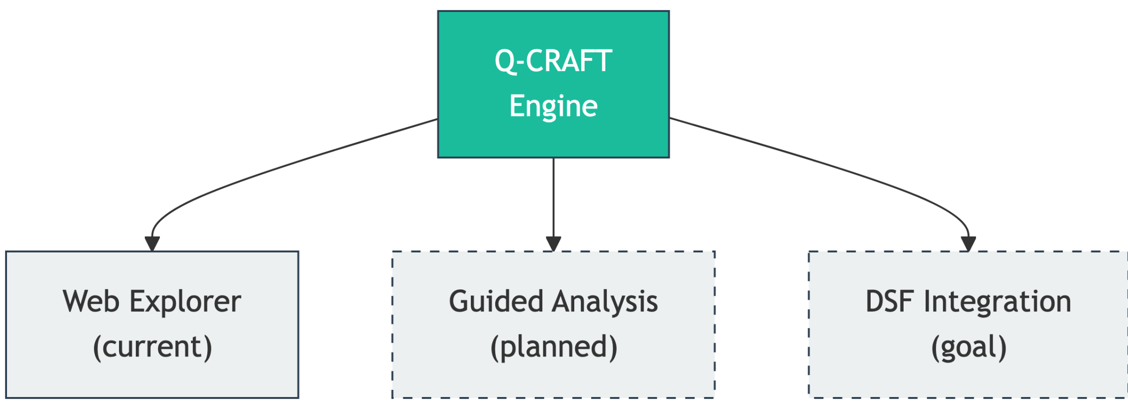
The testing pipeline is public and automated. Baseline parity is exact for 147 of 147 tested countries: zero percentage-point deviation across debt-to-GDP, revenue, primary balance and primary expenditure as shares of GDP, with a further 25 parameter sensitivity combinations passing. Climate-scenario parity is confirmed for ratio metrics, meaning debt-to-GDP, revenue and expenditure as percent of GDP. This kind of transparent, automated verification is essential for tools where the stakes are higher. Chapter 2 has the fuller statement of what the claim covers.

A.3 Q-CRAFT is the proof of concept

Q-CRAFT is deliberately small: five user-facing parameters, one core equation. We chose it because it demonstrates the approach in miniature.

The real goal is the LIC-DSF, the Low-Income Country Debt Sustainability Framework. The LIC-DSF modeling chain (Q-CRAFT, DIGNAD, LIC-DSF) is far more complex, far more consequential, and used for every low-income country. We are already working with end-users of the LIC-DSF, and we want to work with its designers too.

A.4 A modular engine, multiple interfaces



One engine, multiple interfaces

The same engine powers different front-ends for different needs:

1. **Web Explorer (current).** Interactive charts for quick country analysis. Select a country, adjust parameters, compare scenarios.
2. **Guided Analysis Tool (planned).** AI-assisted reasoning through input assumptions. Always available, cheaper than a technical assistance mission. Helps users understand what to enter and why.
3. **DSF Integration (the goal).** Q-CRAFT and DIGNAD feeding into LIC-DSF assumptions, with the same ergonomic layer applied to a far more consequential tool.

The current process makes it hard to do things the right way. Documentation is separated from implementation. If you do not know what to enter for a parameter, you search the user guide and hope it is there (and often it is not). The goal is design that makes it easy to do the right thing and hard to do the wrong thing.

A.5 The co-design invitation

We built V1. Here are specific areas where practitioner input shapes V2.

1. **Parameter guidance.** How should we help users choose productivity growth, expenditure rigidity, and debt target values? What heuristics do experienced practitioners use?
2. **Default assumptions.** What are sensible defaults by country type? Should defaults vary by region?
3. **Capacity development integration.** How would this tool fit into an actual CD workshop or IMF mission?
4. **Validation priorities.** Which countries and scenarios should we prioritize for verification?
5. **Missing features.** What is the first thing you would want that is not here?

What we are asking for: a few hours from a few people for design feedback, continued communication so we build credibly, and feedback on whether we are making something useful.

What we are NOT asking for: official endorsement (we understand institutional constraints), access to confidential information (we work with publicly available materials), or money (this is grant-funded, MIT-licensed open source).

A.6 Why start here

Two reasons to start with Q-CRAFT:

1. **Immediate value.** A co-designed Q-CRAFT tool could be in use within weeks. It costs nothing and requires minimal commitment to evaluate. It complements, not replaces, the Excel workbook.
2. **Pioneer the approach.** Success with Q-CRAFT builds trust and methodology for the LIC-DSF. We prove the design process works on a small tool before applying it to the one that matters most.

A.7 The SovTech vision

Central banks have been applying technology to supervisory functions for years. The BIS documented "SupTech generations" as a framework for technology-driven supervisory and regulatory solutions. The question is no longer whether such tools belong in public financial institutions. The question is when sovereign debt management will catch up.

We call this **SovTech**: the application of SupTech principles to sovereign debt analysis. Three principles guide this work:

Modularity that respects how people work. Same engine, different interfaces. Components like building blocks: swap one module without rebuilding everything.

Open source as institutional infrastructure. Transparency means anyone can examine how analysis is done. MIT license, no barriers to adoption. The calculation engine and the interface are separate concerns, developed and tested independently.

AI as enabler, with a trust layer. The cost of building analytical software has dropped dramatically. But sovereign debt analysis serves policy decisions worth billions of dollars. Verification pipelines, automated testing, and human-in-the-loop governance are requirements, not optional features. This is why we built the parity verification pipeline described in Chapter 2.

Q-CRAFT Explorer is our attempt to show what this looks like. The source code is on GitHub. We welcome your input.

Glossary

Climate scenarios Six warming pathways used in Q-CRAFT, based on IPCC Shared Socioeconomic Pathways (SSPs). The coolest is Paris-aligned (below 2°C warming by 2100, SSP1-2.6). The hottest is Hot Unadapted (high warming with slow adaptation, SSP3-7.0 at the 90th percentile). Climate impacts on GDP are derived from the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on the empirical estimates of Kahn et al. (2021). See the IMF User Guide, Section IV.B for detailed methodology.

C-PIMA (Climate Public Investment Management Assessment) An IMF framework for evaluating how well a country's public investment management institutions account for climate change. C-PIMA assessments include an annex that uses Q-CRAFT to model the fiscal impacts of climate scenarios for the assessed country. See the C-PIMA Handbook (IMF, 2025) for the full framework.

DSA (Debt Sustainability Analysis) A framework used by the IMF and World Bank to assess whether a country's debt is sustainable (that is, whether the government can meet its current and future debt obligations without requiring debt relief or accumulating arrears). Q-CRAFT projections can complement DSA analysis by providing long-term climate-adjusted fiscal trajectories.

Debt dynamics equation The core equation in Q-CRAFT: $d_t = d_{t-1} \times \frac{1+r_t}{1+g_t} - pb_t$. Next year's debt-to-GDP ratio equals the current ratio, adjusted for the interest-growth differential, minus the primary balance. When interest rates exceed growth ($r > g$), debt rises automatically: the government must run a primary surplus even to hold debt still. See the IMF User Guide, Section IV.A on debt dynamics for detailed derivation and corollaries.

Debt-to-GDP ratio Total government debt expressed as a percentage of nominal GDP. The standard measure of a country's debt burden, since it scales debt relative to the economy's ability to service it. Q-CRAFT projects this ratio through 2099 under baseline and climate scenarios.

Expenditure rigidity The degree to which government spending resists downward adjustment when climate shocks reduce GDP. Measured on a 0-1 scale. A value of 1.0 means spending is fully sticky: it stays at the baseline level in local currency terms even as GDP falls, representing the worst case for debt accumulation. A value of 0.0 means spending adjusts fully with GDP, maintaining the baseline expenditure-to-GDP ratio. This parameter only affects climate scenarios, not the baseline. See the IMF User Guide, pp. 20 and 35-36.

Fiscal rule A mechanism in Q-CRAFT that adjusts primary expenditure to steer debt toward a target ratio. When enabled, the model computes the debt-stabilizing primary balance and adjusts spending by the fiscal gap (the difference between the actual and debt-stabilizing primary balance). The adjustment is additive in levels (not a rate) and depends on the prior year's state, requiring recursive year-by-year computation. See the IMF User Guide, pp. 15-18.

DIGNAD (Debt, Investment, Growth, and Natural Disasters) An IMF dynamic general equilibrium model for analyzing the macroeconomic effects of natural disasters and public investment in developing countries. More complex than Q-CRAFT, with substantially more parameters. Discussed in the co-design appendix as an example of a tool that could benefit from the SovTech approach.

Golden master In software verification, a reference output used to detect unintended changes. Q-CRAFT's golden master tests compare the Python engine's outputs against values extracted from the IMF's Excel workbook for the same country and parameter inputs. Parity with the Excel tool (not any independent calculation) is the acceptance criterion.

LIC-DSF (Low-Income Country Debt Sustainability Framework) The IMF-World Bank framework for assessing debt sustainability in low-income countries. Uses country-specific debt burden thresholds based on a composite indicator of institutional quality and macroeconomic fundamentals. Compared against Q-CRAFT in Module 5, and discussed in the co-design appendix as a more complex tool that could benefit from the SovTech approach.

Interest-growth differential The difference between the nominal interest rate on government debt (r) and nominal GDP growth (g), expressed as $(r-g)/(1+g)$. When positive ($r > g$), debt dynamics are unfavorable: the debt ratio rises automatically. When negative ($g > r$), debt dynamics are favorable. This single variable determines whether debt is self-stabilizing or self-reinforcing.

Primary balance Government revenue minus non-interest expenditure (primary expenditure), expressed as a share of GDP. The primary balance is the fiscal variable that a government can control in the near to medium term through fiscal policy. A primary surplus means the government generates enough revenue to cover its non-interest spending. A primary deficit means it does not.

WEO (World Economic Outlook) The IMF's flagship publication providing analysis and projections of the global economy. Q-CRAFT uses the WEO database (currently October 2024 vintage) as the source for macroeconomic and fiscal data through the medium-term forecast horizon (currently through 2029). Beyond this horizon, Q-CRAFT's own projection methodology takes over.

SovTech The application of SupTech (supervisory technology) principles (modularity, open source, human-centered design) to sovereign debt analysis and public financial management tools. Q-CRAFT Explorer is a proof of concept for this approach.

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